MATHEMATICS GRADE 10: VECTORS I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 3.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 3.0: Geometry
Sub-Strand	Sub-Strand 3.3: Vectors I
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Definition of a vector and distinction between scalar and vector quantities – Representation of vectors: column vectors, position vectors, and directed line segments – Magnitude (modulus) of a vector and its calculation – Addition and subtraction of vectors geometrically and algebraically – Multiplication of a vector by a scalar – The zero vector and negative of a vector – Parallel vectors and collinear points – Translation vectors and their application in the coordinate plane – Mid-point of a vector and section formula – Expressing vectors in terms of given vectors (vector paths)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) define a vector and distinguish it from a scalar quantity in real-life Kenyan contexts b) represent vectors as column vectors, position vectors, and directed line segments on a Cartesian plane c) perform addition, subtraction, and scalar multiplication of vectors both geometrically and algebraically d) determine the magnitude of a vector and identify parallel and collinear vectors e) apply vectors to solve problems involving translations, mid-points, and real-life displacement situations such as navigation across Lake Victoria or movement on Kenyan road networks
Core Competencies	– Communication and Collaboration: Learners discuss and justify vector operations in pairs and groups, presenting solutions to peers during class activities and DQB sessions – Critical Thinking and Problem Solving: Learners analyse vector paths, determine resultant displacements, and identify collinear points by constructing logical, multi-step mathematical arguments – Creativity and Imagination: Learners design vector diagrams and physical models that represent real-world navigational or structural scenarios such as ferry routes on Lake Victoria – Digital Literacy: Learners engage with GeoGebra simulations and online resources to visualise vector addition, scalar multiplication, and translations dynamically – Self-Efficacy: Learners build confidence through progressive problem difficulty, moving from guided examples to independent vector proofs and applied tasks

	– Learning to Learn: Learners reflect on misconceptions about direction and magnitude, update their Driving Question Board, and revise cumulative vector models across lessons
Core Values	– Responsibility: Learners take ownership of their vector model revisions and ensure accuracy in representing real-world displacement problems affecting communities (e.g., ferry navigation safety on Lake Victoria) – Respect: Learners appreciate diverse approaches to vector problems during group discussions, listening actively to peers' geometric and algebraic reasoning – Unity: Learners collaborate across ability groups to solve vector path problems, reflecting the spirit of teamwork seen in Kenyan community development projects – Social Justice: Learners connect vector concepts to equitable resource distribution planning (e.g., mapping shortest delivery routes to remote Kenyan counties) – Integrity: Learners show honest self-assessment during model-building phases, accurately acknowledging errors and correcting their vector diagrams
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners formulate inquiry questions about why direction matters in navigation and sport, anchoring the need for vectors versus scalars – Developing and Using Models: Learners iteratively build and revise arrow-based vector diagrams and coordinate-plane models to represent displacement scenarios throughout the unit – Planning and Carrying Out Investigations: Learners measure and record displacement vectors physically (using rulers and protractors on grid paper) to investigate vector addition by the triangle and parallelogram rules – Analysing and Interpreting Data: Learners compare calculated resultant vectors with measured diagrams, identifying sources of error and verifying algebraic results geometrically – Using Mathematics and Computational Thinking: Learners apply column vector notation, magnitude formulae, and scalar multiplication systematically to solve multi-step problems – Constructing Explanations and Designing Solutions: Learners use vector reasoning to explain navigational routes, structural force directions, and mid-point locations in applied Kenyan contexts – Engaging in Argument from Evidence: Learners justify claims about parallel vectors and collinear points using vector algebra, defending conclusions with mathematical evidence during class debates
Pertinent & Contemporary Issues (PCIs)	– Safety: Learners are reminded to use rulers and compasses carefully during practical vector drawing activities to avoid injury – Life Skills — Decision Making: Learners apply vector reasoning to make optimal routing decisions, such as the most efficient path for a matatu across Nairobi's road network – Life Skills — Problem Solving: Learners use vector addition to resolve competing displacement directions, mirroring real decisions faced by pilots, sailors, and engineers in Kenya – Environmental Education: Learners connect vector displacement to conservation mapping, such as tracking wildlife migration corridors in the Maasai Mara using directional data – Career Guidance: Learners explore how vectors underpin careers in engineering, aviation, architecture, game development, and geospatial mapping — all growing sectors in Kenya's Vision 2030 economy
Career Connections	– Civil and Structural Engineer: Vectors are used to analyse forces on bridges, buildings, and infrastructure projects such as the Nairobi Expressway – Pilot and Aviation Professional: Vectors model aircraft velocity, wind correction, and navigation headings for routes across Kenyan airspace – Marine Navigator / Ferry Operator: Vectors calculate resultant displacement for vessels crossing Lake Victoria between Kisumu, Mfangano, and Mbita – Geospatial Analyst / Surveyor: Vectors represent land boundaries, GPS coordinates, and displacement in mapping projects

	across Kenya – Game Designer / Software Developer: Vectors control character movement, physics engines, and camera direction in digital games — a booming field in Nairobi's Silicon Savannah – Architect and Urban Planner: Vectors describe structural load directions and spatial layouts in housing and city planning projects in Nairobi and Kisumu
Focus for Lessons	This unit introduces Grade 10 learners to the foundational concepts of vectors, emphasising the critical distinction between quantities that have magnitude only (scalars) and those that have both magnitude and direction (vectors). Using the anchoring phenomenon of a ferry crossing Lake Victoria — where wind, current, and engine thrust must all be combined to reach the correct destination — learners progressively build understanding of vector representation, operations, and applications. The unit follows a predict–observe–explain–model cycle across 8 lessons, grounding abstract algebraic operations in vivid Kenyan navigational, agricultural, and engineering contexts.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why did the Kisumu ferry miss Mbita — and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind? KICD KEY INQUIRY QUESTIONS: 1. How do vectors differ from scalars, and why is direction essential in describing some real-world quantities? 2. How can vectors be represented, combined, and applied to solve problems involving displacement, navigation, and geometry?
Anchoring Phenomenon	ANCHORING PHENOMENON — The Drifting Ferry on Lake Victoria A passenger ferry departs from Kisumu port heading directly east toward Mbita, a straight-line distance of approximately 60 km. The captain sets the engine to push the vessel due east at 20 km/h. However, a strong southward current of 8 km/h is also acting on the ferry at the same time. When the ferry arrives, it has drifted significantly south of Mbita and lands at the wrong point on the shore, causing delay and confusion for hundreds of passengers and traders. Students observe this scenario via a vector diagram and are immediately puzzled: if the engine was pointing directly at Mbita, why did the ferry miss? How can two separate pushes — one east and one south — be combined into a single resultant journey? How far did the ferry actually travel, and in what true direction? Why does the order of the pushes not matter, but the direction of each push matters enormously? This phenomenon is puzzling because learners' intuition suggests the ferry should travel east if the engine points east — they have not yet encountered the idea that multiple simultaneous directed quantities combine into a single resultant vector. It creates an immediate need for the mathematical tools of the entire unit: vector representation, addition, magnitude calculation, and direction — making every lesson feel purposeful and necessary.
Storyline Thread	Lesson 1: Scalars vs. Vectors — Introducing the Ferry Problem Learners encounter the Kisumu–Mbita ferry anchoring phenomenon for the first time. They predict why the ferry drifted and debate whether speed alone is enough information to describe the journey. The class constructs an initial Driving Question Board (DQB) and learners are introduced to scalar vs. vector quantities using Kenyan examples (temperature in Kericho = scalar; wind from Mombasa to Nairobi = vector). Learners make their first vector arrows on grid paper. Lesson 2: Representing Vectors — Column Vectors and Position Vectors Learners learn to represent vectors using directed line segments, column vector notation, and position vectors on the Cartesian

	plane. They map the ferry's engine thrust and the lake current as column vectors and place them on a coordinate grid. The concept of equal vectors and the negative of a vector is established. Learners revise the DQB with new representational tools. Lesson 3: Vector Addition — Triangle and Parallelogram Rules Learners physically add the ferry's engine vector and current vector using the triangle rule on grid paper, discovering the resultant displacement. They verify results using the parallelogram rule and connect both methods algebraically using column vector addition. The class recalculates where the ferry actually landed and updates the DQB. Lesson 4: Subtraction of Vectors and the Zero Vector Learners explore what it means to subtract a vector — framed as: "What correction vector must the captain apply to return to Mbita from where the ferry landed?" They define the zero vector, the negative vector, and practice subtraction algebraically and geometrically. Supporting phenomenon: the Rift Valley farmer's rope cart is revisited for subtraction context. Lesson 5: Scalar Multiplication of Vectors and Parallel Vectors Learners multiply vectors by scalars, discovering that this changes magnitude but not direction (or reverses it for negative scalars). They identify parallel vectors and discuss what parallel means in a vector context — connecting to the drone surveillance scenario in Kericho. Learners update their cumulative vector model on the DQB. Lesson 6: Magnitude of a Vector — How Far Did the Ferry Really Travel? Learners apply Pythagoras' theorem to calculate the magnitude (modulus) of a vector from its column vector components. They calculate the true distance the ferry travelled and compare it to the intended 60 km route. The concept of a unit vector is introduced. Learners revise their ferry displacement model with correct magnitudes labelled. Lesson 7: Collinear Points and Vector Paths Learners use vectors to prove that three points are collinear — framed as: "Were the ferry's launch point, drift point, and Mbita port ever on the same straight line?" They practice expressing vectors in terms of given vectors along geometric paths (e.g., triangles and quadrilaterals using Kenyan landmark maps). The DQB is updated with collinearity evidence. Lesson 8: Translation Vectors and Unit Synthesis — Solving the Ferry Problem In the final lesson, learners apply translation vectors to map the complete corrected ferry route — calculating the exact direction and speed adjustment the captain needs to compensate for the current and arrive precisely at Mbita. Learners complete their cumulative vector model, deliver a short group presentation answering the driving question, and reflect on how vectors underpin careers in navigation, engineering, and geospatial mapping in Kenya.
Supporting Phenomena	– A Nairobi matatu takes a diagonal shortcut across a grid of streets: learners puzzle over why the shortcut distance and direction cannot simply be described by a single speed, launching the need for column vectors and magnitude – A farmer in the Rift Valley pulls a water cart with two ropes at different angles — the cart moves in a direction neither rope is pointing, illustrating vector addition by the parallelogram rule – A 100 m sprinter at the Nyayo National Stadium runs north, then east — arriving at a different point than a sprinter who ran the same distances in the opposite order (east then north), provoking curiosity about vector paths and whether order matters (it does not for displacement — reinforcing commutativity of vector addition) – A drone used for crop surveillance in a Kericho tea estate flies with a velocity vector that combines its own engine thrust and the wind vector — the farmer on the ground observes the drone drifting off its planned path, connecting to resultant vectors

	and scalar multiplication for course correction

LESSON 1 (40 minutes): Scalars vs. Vectors — Launching the Ferry Problem	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To introduce learners to the distinction between scalar and vector quantities by anchoring the concept in the real-world puzzle of why the Kisumu–Mbita ferry drifted off course, creating a compelling need for the entire unit.
Knowledge	Learners will know that scalar quantities have magnitude only, that vector quantities have both magnitude and direction, and that direction is essential for describing quantities such as displacement, velocity, force, and current.
Skills	Learners will be able to classify everyday Kenyan quantities as scalars or vectors, represent a vector as a directed arrow on grid paper, and articulate why the ferry missed Mbita using directional language.
Attitudes	Learners will show curiosity when confronting a counterintuitive phenomenon, willingness to revise initial predictions, and appreciation for mathematics as a tool that solves real navigation problems in Kenya.
Key Inquiry Question	How do vectors differ from scalars, and why is direction essential in describing some real-world quantities?
Purpose in Storyline	This is the opening lesson that launches the anchoring phenomenon. Learners encounter the drifting ferry for the first time, form initial predictions, surface prior knowledge about quantities, and open the Driving Question Board. Every subsequent lesson will return to this phenomenon to deepen the mathematical model.
Safety Notes	No hazardous materials are used. Ensure grid paper and pencils are distributed before the lesson begins to avoid disruption. If learners express anxiety about the ferry scenario relating to personal experiences near Lake Victoria, acknowledge their feelings calmly and keep the focus on the mathematical puzzle.

B. LESSON OVERVIEW

Learners watch or hear a vivid narrative of the Kisumu–Mbita ferry drifting south due to a lake current, then predict why this happened using only their prior intuition. Through structured discussion, they surface the inadequacy of speed alone and discover that direction matters. The teacher introduces the terms scalar and vector using familiar Kenyan contexts — temperature in Kericho, wind from Mombasa, matatu distance from Nairobi to Nakuru — before learners draw their first vector arrows on grid paper representing the engine thrust and the current. The lesson closes with the class opening a Driving Question Board populated with their burning questions, which will drive the unit forward.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Scalar multiplication: component form] Source: Khan	Learners listen to the teacher narrate the ferry scenario without any diagram: a ferry leaves Kisumu heading due east toward Mbita, 60 km away, engine pushing at 20 km/h due east. A	Narrate slowly and dramatically: 'Imagine you are a trader in Mbita. You have been waiting since morning for goods arriving on the Kisumu ferry. The ferry left on time, the engine was running	Elicitation of prior knowledge and prediction: learners activate intuitive ideas about motion and direction, making their thinking visible so the teacher can identify and later address the core	Teacher reads sticky notes or listens to predictions to identify the dominant misconception — likely that the ferry should have stayed on course because the engine pointed east. This informs the

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX) Source: CK-12 🔍 Search ARES for similar readings	southward lake current of 8 km/h is also acting on the ferry. The ferry arrives at the wrong place. Learners individually write on a sticky note or in their exercise books: (1) Where do you think the ferry ended up — still at Mbita, north of Mbita, or south of Mbita? (2) Why did this happen? They then share predictions in pairs before a brief whole-class share-out. Most learners will predict the ferry stayed on course because the engine pointed east, creating productive confusion.	perfectly, the captain was experienced. But when the ferry arrives — it is not at Mbita. It has landed far to the south. Your goods are stranded. Hundreds of passengers are confused. What went wrong?' Pause for 10 seconds of silence after the narrative before asking learners to write their predictions. Do not correct any prediction at this stage. Circulate and read sticky notes silently. After pair discussion, ask two or three learners to share: 'Tell us your prediction and your reason.' Accept all answers neutrally: 'Interesting — let us keep that idea on the table.' WAIT TIME: After each learner shares, wait 5 seconds before responding or moving to the next learner, signalling that thinking time is valued.	misconception that speed alone determines destination.	depth of contrast needed in the Observe Phase. A quick show of hands — 'How many predicted it landed south?' versus 'How many predicted it stayed at Mbita?' — gives a class-wide picture.
Observe Phase 📹 **VIDEO:** Scalar multiplication: component form Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX) Source: CK-12 🔍 Search ARES for similar readings	The teacher reveals a large, simple vector diagram drawn on the board or projected: a grid with Kisumu on the left, Mbita on the right along a horizontal line, an eastward arrow labelled 'engine: 20 km/h east', and a downward arrow labelled 'current: 8 km/h south'. A dashed diagonal arrow shows where the ferry actually went — south-east of the intended path — landing at a point clearly south of Mbita. Learners observe the diagram for 60 seconds in silence, then discuss in groups of three: 'What do you notice? What surprises you? What changed your mind about your prediction?' Groups share observations. The teacher draws attention to the two arrows — each has a length (magnitude) and a direction — and asks: 'What if I told you only the speed of each push, without the	Display the diagram and say: 'Look at this carefully for one full minute. Do not talk yet.' After 60 seconds: 'Now turn to your group — what do you notice that you did not expect?' WAIT TIME: Give groups 2 minutes of uninterrupted talk time before calling on anyone. When debriefing, ask: 'Who noticed something their neighbour did not?' To sharpen the key insight, point to one arrow and say: 'If I erase the arrowhead and just leave the line, what information have I lost?' Pause for 8 seconds. Learners should arrive at: direction. Confirm: 'Yes — the arrowhead carries information that a plain number cannot. That is the heart of today's lesson.'	Observation of a phenomenon diagram followed by structured talk: learners move from intuition to evidence by comparing what they predicted with what the diagram shows, surfacing the role of direction as distinct from magnitude.	Listen for whether learners spontaneously use directional language — 'south', 'downward', 'the current pushed it away'. If direction language is absent, probe: 'You said the ferry drifted — drifted which way?' This checks whether learners are beginning to distinguish directional from non-directional description.

	arrow direction — would you be able to draw this diagram?' Learners respond no, creating the conceptual need for direction.			
Explain Phase 📹 **VIDEO:** [Scalar multiplication: component form] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	The teacher formally introduces scalar and vector quantities using three paired Kenyan examples: (1) The temperature in Kericho tea farms today is 18 degrees Celsius — this is a scalar, magnitude only, no direction needed. The wind blowing from the Indian Ocean toward Nairobi is 25 km/h from the south-east — this is a vector, it needs both size and direction. (2) The distance from Nairobi to Nakuru along the highway is 160 km — scalar. The displacement of a matatu travelling directly from Nairobi to Nakuru is 160 km due north-west — vector. (3) The mass of a sack of maize from Eldoret is 90 kg — scalar. The force a porter exerts pushing that sack up a ramp at Mombasa port is 200 N directed up the slope at 30 degrees — vector. Learners complete a T-chart in their books labelled 'Scalar' and 'Vector' and classify eight additional quantities provided on a handout or written on the board: speed, velocity, time, force, energy, weight, temperature, displacement. Learners then draw their first two vector arrows on 1 cm grid paper: Arrow A pointing 4 units east labelled 'engine thrust' and Arrow B pointing 2 units south labelled 'lake current'. They label each arrow with its magnitude and direction.	Introduce the formal definition: 'A scalar quantity is fully described by a magnitude alone. A vector quantity requires both a magnitude and a direction to be fully described.' Write both definitions on the board and leave them visible for the rest of the lesson. For each Kenyan example, first present the scalar version and ask: 'Is this complete? Can you act on this information alone?' Then present the vector version: 'Now what extra information do you have?' For the T-chart activity, circulate and look for misclassifications — particularly weight (vector, because it acts downward) versus mass (scalar). Do not correct immediately: 'Interesting choice — tell me your reasoning.' After learners draw their arrows, select two volunteers to draw their arrows on the board. Ask the class: 'Are these two arrows the same? What makes an arrow a vector and not just a doodle?' WAIT TIME: After posing this question, wait 10 seconds before accepting responses, allowing deeper processing.	Concept introduction through contrasting cases: by presenting scalar and vector versions of the same Kenyan context side by side, learners experience the insufficiency of magnitude alone and construct the need for direction as a mathematical object. The grid-paper arrow activity bridges the conceptual and representational.	Collect or scan T-charts to check classification accuracy. The critical diagnostic item is weight versus mass — if most learners classify both as scalars, address the distinction explicitly in Lesson 2. Check that vector arrows on grid paper have both an arrowhead (direction) and a label (magnitude). Missing arrowheads indicate the direction concept has not yet been internalised.
Driving Question Board (DQB) Creation 📹 **VIDEO:**	Each learner receives two sticky notes (or writes in a designated section of their book). On the first sticky note they write one question the ferry phenomenon makes them	Give learners 3 minutes of quiet individual writing time before any sharing. When posting questions, read each one aloud without evaluating it: 'Someone asks —	Question generation and public display: making learner questions visible transforms the class into a community of inquiry. The DQB externalises curiosity and creates	Analyse the DQB questions for depth and focus. If most questions are about the ferry story rather than the mathematics (for example, 'Why does the lake have

DNA as the "transforming principle" Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX) Source: CK-12 🔍 Search ARES for similar readings	want to answer — something they genuinely do not yet know. On the second they write one thing they think they already know that might help explain the ferry problem. The teacher collects the first-note questions and sticks them on a large sheet of paper or section of the board labelled 'Our Driving Question Board'. Questions are grouped loosely by theme as they are posted — 'questions about direction', 'questions about combining pushes', 'questions about distance'. The class reads the DQB together and the teacher frames it: 'These are the questions our unit of work on Vectors will answer. Every lesson, we will return here and mark off what we now understand.' The teacher adds the driving question at the top: 'Why did the Kisumu ferry miss Mbita — and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind?'	how do you add two pushes together? Great question — that goes under combining vectors.' If a question seems off-topic, affirm the curiosity: 'That is a fascinating question — let us put it in a parking lot corner and revisit it.' After the DQB is assembled, step back and say: 'Look at what this class wants to know. These are real mathematical questions that engineers, pilots, and sailors in Kenya use every day.' Photograph or preserve the DQB — it will be updated in every subsequent lesson. WAIT TIME: After the class reads the full DQB silently, wait 15 seconds before speaking, allowing learners to notice patterns and connections in the questions.	accountability — learners will see their own questions answered as the unit progresses, giving each lesson a felt purpose.	currents?'), probe gently: 'What mathematical tools would you need to answer your own question?' If questions already reference addition, angle, or distance, note these as indicators of high prior knowledge to build on in Lessons 3 and 6.
Model Building Phase 📹 **VIDEO:** Scalar multiplication: component form Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX) Source: CK-12	Learners return to their grid paper and, working in pairs, attempt a first rough model of the ferry journey using only what they now know. They draw Kisumu on the left of a grid, Mbita 6 grid squares to the right (representing 60 km at 10 km/h per square), draw Arrow A (4 squares east, the engine), and Arrow B (2 squares south, the current, drawn starting from the tip of Arrow A to show the sequential pushes — tip-to-tail). They mark the endpoint and label it 'Where the ferry actually landed'. They annotate their diagram: 'The ferry missed Mbita because both magnitude AND direction had to be considered.' Pairs share their	Circulate during pair work and ask probing questions rather than giving answers: 'How did you decide how long to make Arrow B?', 'What does the endpoint of your second arrow represent?', 'If the current were faster — say 16 km/h instead of 8 — what would change on your diagram?' These questions seed thinking for Lessons 3 and 6. Do not introduce the term resultant yet — allow learners to describe the combined arrow in their own language. If a pair draws both arrows starting from the same point (parallelogram style), affirm this as also valid: 'Interesting — you found a different way to show the same thing. We	Preliminary model construction: learners externalise their current understanding in a visual-spatial form, making implicit reasoning about direction and combination explicit. The incompleteness of the model is intentional — it creates productive cognitive tension that motivates subsequent lessons.	Review pair diagrams for three key indicators: (1) Both arrows are present and directed. (2) The arrows are drawn tip-to-tail, placing the ferry endpoint south of Mbita. (3) The learner annotation references direction as the cause of the drift, not only speed. Pairs who draw only one arrow or place the endpoint at Mbita still hold the core misconception and need targeted attention in Lesson 2 when column vectors are introduced.

🔍 Search ARES for similar readings	diagrams with another pair and compare. The teacher closes by framing what is still unknown: 'You have drawn where the ferry went — but do you know exactly how far it travelled? Or the exact angle it drifted? That is what the next seven lessons will give you the tools to find.'	will explore that method in Lesson 3.' Close the model-building phase with: 'Your diagram is a mathematical model. It is incomplete — but it is the beginning of the answer to our driving question.' WAIT TIME: After the closing statement, pause for 8 seconds to let the framing land before dismissing.		

D. TEACHER REFLECTION

After the lesson, consider: Did the ferry narrative create genuine curiosity or did it feel forced? Were learners able to articulate why direction matters using their own words, or did they reproduce the teacher definition? Were the Kenyan scalar-vector examples culturally grounding or did any feel unfamiliar — if Kericho or Mombasa port examples were unfamiliar, substitute with local equivalents next time. Did the DQB questions reveal conceptual readiness for vector addition earlier than expected in some learners? If so, plan a brief extension in Lesson 3. Most importantly: did learners leave the room wanting to know the answer to the driving question? If not, the phenomenon needs to be made more personal — consider asking whether any learner has taken the Kisumu–Mbita ferry or has family who use Lake Victoria transport routes.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A passenger ferry departed Kisumu heading due east toward Mbita but arrived far south of the intended destination because a southward lake current was also acting on the vessel at the same time.
What did I learn?	Quantities that have only magnitude are called scalars. Quantities that have both magnitude and direction are called vectors. Direction is not optional information — it is mathematically essential for describing motion, force, and displacement.
How does this explain the phenomenon?	The ferry missed Mbita because the engine thrust (a vector pointing east) and the lake current (a vector pointing south) both acted simultaneously. Since direction matters, the two pushes combined to produce a resultant journey that was neither purely east nor purely south — but diagonally south-east, landing the ferry at the wrong point on the shore.

LESSON 2 (80 minutes): Representing Vectors — Column Vectors and Position Vectors	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to represent vectors using column notation and position vectors, and to connect these representations to real displacement on a coordinate grid — building the mathematical language needed to describe the ferry's drift on Lake Victoria.
Knowledge	Learners will know that a vector can be written as a column vector with horizontal and vertical components, and that a position vector describes the location of a point relative to the origin.
Skills	Learners will be able to write vectors in column notation, identify position vectors of points on a Cartesian plane, draw vectors on a grid, and describe the displacement of the Kisumu ferry in column vector form.
Attitudes	Learners appreciate precision and accuracy in mathematical representation, recognising that a single missing detail — such as direction — can cause real-world consequences like the ferry missing Mbita.
Key Inquiry Question	How can vectors be represented, combined, and applied to solve problems involving displacement, navigation, and geometry?
Purpose in Storyline	In Lesson 1, learners distinguished vectors from scalars and understood that the ferry drifted because direction matters. In this lesson, learners gain the precise mathematical tool — column vectors and position vectors — to write down exactly what happened to the ferry: an eastward component and a southward component expressed as a single column vector. This representation is the foundation for vector addition, magnitude, and navigation corrections in later lessons.
Safety Notes	No physical hazards. Ensure learners use rulers when drawing vectors on grids to maintain accuracy. Remind learners that neatness and labelling of diagrams are mathematical safety nets — an unlabelled arrow has no mathematical meaning.

B. LESSON OVERVIEW

Learners revisit the Kisumu–Mbita ferry scenario and ask: now that we know direction matters, how do we write a vector down precisely? Through a structured predict–observe–explain sequence, learners discover column vector notation by decomposing the ferry's engine thrust and the lake current into horizontal and vertical components on a coordinate grid. They then extend this to position vectors, locating Kisumu port and Mbita on a Cartesian plane and describing the ferry's intended and actual positions as position vectors. The lesson uses grid-based drawing activities, a guided worked-example gallery walk, and a collaborative whiteboard sensemaking task. By the end, every learner can express the ferry's eastward thrust as (20, 0) and the southward current as (0, −8) — and can see these two column vectors sitting inside the same diagram that puzzled them in Lesson 1.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan	Learners are shown a large printed or projected Cartesian grid with Kisumu port marked at the origin (0, 0). The teacher asks: 'Mbita is 6 grid squares to the east. Write	1. Display the grid with Kisumu at the origin. Say: 'Here is Kisumu port. Mbita is directly east — 6 squares away on this grid. I want you to write down the	Prediction Wall — learners' unfiltered prior ideas are displayed publicly. This strategy surfaces informal notations and creates cognitive dissonance when	Observable indicator: Can the learner write any directional instruction that includes both a magnitude and a sense of direction (even informally)? Red

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	down, in any way you like, the instruction you would give to a robot to move from Kisumu to Mbita using only numbers.' Learners work individually for 3 minutes in their exercise books, then share with a shoulder partner. The class hears 4–5 responses read aloud. Most learners write '6 steps east' or simply '6'. A few may write (6, 0) by intuition. The teacher records all responses on the board without correction, creating a visible prediction wall. Learners are then told: 'By the end of today, you will have one agreed, powerful way to write any vector — and you will use it to describe exactly what happened to the ferry.'	mathematical instruction to get from Kisumu to Mbita. Use any notation you like — there is no wrong answer yet.' 2. Allow 3 minutes of individual silent writing. Do NOT circulate or hint. 3. After 3 minutes, say: 'Share your idea with your shoulder partner. You have 90 seconds.' 4. Cold-call exactly 5 learners using name sticks. Record every response verbatim on the board. Likely responses: '6 east', '6', '+6', '(6, 0)', 'six steps right'. 5. Say: 'Look at all these ideas. Some of you wrote just a number. Some wrote a number and a direction. We are going to find out today which of these is the most powerful for solving the ferry problem — and why.' 6. Leave all predictions visible throughout the lesson.	learners later see that '6' alone is insufficient to describe the ferry's drift, while (6, 0) and (0, −8) together tell the full story. It gives every learner a personal stake in the lesson's resolution.	flag: learners who write only '6' with no directional component — these learners still conflate scalar and vector thinking from Lesson 1 and will need targeted support during Phase 3.
Observe Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a structured activity sheet with three tasks. Task A: A pre-drawn Cartesian grid shows four labelled arrows (vectors AB, CD, EF, GH) going in different directions with different lengths. Learners measure the horizontal and vertical 'steps' of each arrow using the grid squares, recording their readings in a table with columns labelled 'Horizontal steps (right = +, left = −)' and 'Vertical steps (up = +, down = −)'. Task B: Learners observe that the teacher writes the two readings for vector AB as a column: the horizontal reading on top and the vertical reading below, enclosed in a bracket — the column vector. They copy this for all four vectors. Task C: The ferry scenario is returned to. Learners are shown: arrow 1 — the engine thrust, 6 squares east; arrow 2 — the lake current, 2.4 squares south	1. Distribute activity sheets face-down. Say: 'Do not turn over until I say go.' 2. Say: 'Turn over. Look only at Task A. Using your ruler, count the horizontal grid squares each arrow moves — right is positive, left is negative. Then count the vertical squares — up is positive, down is negative. Write both numbers in the table. You have 5 minutes. Go.' 3. Circulate. Look specifically for: (a) learners who count diagonal distance rather than horizontal/vertical steps — redirect with 'Count only the flat steps first, then only the up-down steps'; (b) sign errors on arrows going left or down. 4. After 5 minutes, cold-call 2 learners to share their table rows. Confirm correct values. 5. Say: 'Now watch.' Live-demonstrate on the board writing vector AB as a column vector. Say: 'The top number tells us left or right. The	Component Decomposition Grid Activity — learners physically count horizontal and vertical steps before they see the column vector formula. This inductive approach means learners construct the meaning of each entry in the column vector from direct observation of directed arrows on a grid, rather than receiving the notation as an abstract rule. Connecting immediately to the ferry's engine thrust and current makes the notation feel like a discovery rather than a definition.	Observable indicators: (1) Learner's table in Task A shows correct sign conventions for all four vectors. (2) Learner writes the ferry engine thrust correctly as a column vector with 6 on top and 0 below, and the current as 0 on top and a negative number below. Red flag: a learner who writes (6, 8) as a single column vector for the ferry's combined motion has jumped ahead but confused resultant with components — note this learner for Phase 3 discussion.

	(scaled). They record both as column vectors. A short 4-minute teacher-led demonstration video (or live board drawing) shows a Kenyan navigator on Lake Victoria decomposing a boat's motion into north–south and east–west components using a grid — reinforcing that this is a real navigational tool, not an abstract exercise.	bottom number tells us up or down. Together they give us the full vector. This is called a column vector.' Write the definition on the board. Allow learners to copy. 6. Say: 'Now complete Task B for all four vectors. 3 minutes.' WAIT. 7. Cold-call 3 learners for Task B answers. Correct misconceptions immediately. 8. Transition to Task C. Say: 'Now go back to our ferry. The engine pushes the ferry 6 squares east. What is the column vector for the engine thrust? Discuss with your partner — 30 seconds.' Cold-call 1 pair. Confirm (6, 0) written as a column. Say: 'The current pushes 2.4 squares south. What is its column vector?' WAIT 10 seconds. Cold-call 1 learner. Confirm (0, −2.4) as a column. 9. Play or draw the navigator demonstration. Pause to ask: 'What are the two components the navigator is separating?' WAIT 15 seconds. Take 2 hands.		
Explain Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners extend column vectors to position vectors. The teacher places Kisumu port at origin O(0,0) on a large class grid. Three points are marked: A = Mbita at (6, 0), B = the ferry's actual landing point at (6, −2.4), and C = a supply depot at (3, 4). Learners are asked: 'How do we describe the position of each of these places as a vector from the origin?' Through guided questioning, learners see that the position vector of point A is simply the column vector from O to A — the same notation they just learned. They write position vectors OA, OB, and OC. Learners then work in pairs on a 'Nairobi Grid' extension problem: five Kenyan cities are plotted on a	1. Draw the class grid with O, A, B, C labelled. Say: 'Point A is Mbita. Its coordinates are (6, 0). I want to describe the vector from O — the origin, Kisumu — to A, using column vector notation. What goes on top? What goes below?' WAIT 10 seconds. Cold-call 1 learner. Write the column vector on the board. Label it 'Position vector of A' and write OA with a vector arrow above it. 2. Repeat the process for B — the actual landing point — asking: 'What does a negative bottom number tell us about where the ferry ended up?' WAIT 12 seconds. Cold-call 2 learners. Guide toward: 'It landed south of where it was meant to go.' 3. Say: 'Write the position vector of	Anchored Concept Extension — learners first master column vectors for free vectors (Task A/B), then the teacher explicitly bridges to position vectors using the same grid and same ferry scenario. By keeping the context constant while extending the concept, learners see position vectors as a natural application of what they already know, not a new unrelated idea. The Nairobi Grid task then generalises the concept to a broader Kenyan context, building transferability.	Observable indicators: (1) Learner correctly writes position vectors OA, OB, OC with correct signs. (2) In the Nairobi Grid task, learner identifies the parallel position vectors by comparing ratios of components, not just by visual inspection. (3) Learner can articulate in one sentence why the ferry's intended position vector has a zero in the bottom row. Red flag: learner who writes position vectors with reversed entries (vertical component on top) needs immediate individual correction — this error will propagate into all subsequent vector work.

	Cartesian grid (Nairobi at origin, Mombasa, Kisumu, Nakuru, Eldoret at various coordinates). Pairs write the position vector of each city and identify which two cities have position vectors that are parallel. The class reconvenes. The teacher facilitates a whole-class explanation of why position vectors are useful for navigation: if you know the position vector of your destination, you know exactly how to get there from the origin — directly answering the driving question about what tool the captain needs.	C in your books. 1 minute.' WAIT. Cold-call 1 learner. 4. Distribute the Nairobi Grid paired task. Say: 'With your partner, write the position vector of each city and find the two cities whose position vectors are parallel. You have 8 minutes. I will be circulating — be ready to explain your reasoning, not just your answer.' 5. Circulate. Ask probing questions: 'How do you know two vectors are parallel just by looking at their column vectors?' Do not answer — redirect to: 'Think about what parallel means on the grid.' 6. After 8 minutes, cold-call 3 pairs to share. Facilitate class agreement on which cities are parallel and why. 7. Anchor back to phenomenon: 'So if the captain had written down the position vector of Mbita before leaving Kisumu, what would it look like? Would it include any southward component?' WAIT 10 seconds. Take 3 hands. Affirm: 'No southward component — the intended position vector is (6, 0). But the current added (0, −2.4) to the journey. We now have the language to describe both precisely.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Intro to the coordinate plane] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:**	Each learner receives two sticky notes. On the first (green), they write one thing they can now say precisely about the ferry's journey using column vectors that they could not say in Lesson 1. On the second (yellow), they write one new question that today's lesson has raised for them — for example: 'Can I add the two column vectors together to find the ferry's actual path?' or 'Does the order in which I write the components matter?' Sticky notes	1. Distribute two sticky notes per learner. Say: 'Green note: write one precise statement about the ferry using column vector language. Yellow note: write one question today's lesson has opened up for you. You have 3 minutes. Work silently.' 2. Circulate and read over shoulders. Identify 3–4 high-leverage sticky notes to read aloud — prioritise notes that (a) correctly use column vector notation and (b) anticipate vector addition. 3. Invite learners to place	Two-Colour DQB Protocol — using green for evidence and yellow for new questions makes visible both the growth in understanding and the productive uncertainty that drives the storyline forward. Publicly reading selected notes validates learner thinking, models precise mathematical language, and creates anticipation for Lesson 3. The teacher's bridge statement explicitly connects today's evidence to the next lesson's content.	Observable indicators: (1) Green sticky note contains a column vector written in correct notation (bracket, two entries, correct signs) applied to the ferry scenario. (2) Yellow sticky note poses a question that is mathematically forward-looking (e.g., combining vectors, finding total distance, finding direction). Red flag: learner whose green note is still written in prose without column vector notation has not yet consolidated the lesson's core

Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	are placed on the class Driving Question Board under two columns: 'Evidence We Now Have' and 'Questions Still Open'. The teacher reads 3–4 sticky notes aloud, making connections visible: 'Notice that Amara says she can now write the current as (0, −8). And Omondi is asking whether we can add (20, 0) and (0, −8). Those two ideas are going to meet in Lesson 3.' The teacher updates the board's central driving question card to show the column vector representation of the ferry's engine and current.	their notes on the DQB. Read the selected notes aloud. Say: 'Listen to how our language has changed since Lesson 1. In Lesson 1 we said the ferry drifted south. Today we can say the current is the column vector (0, −8) and the engine thrust is (20, 0). That is a huge leap in precision.' 4. Point to the 'Questions Still Open' column. Say: 'Many of you are asking how to combine these two vectors. That is exactly where we are going next. Hold that question.'		learning and needs follow-up before Lesson 3.
Model Building Phase 📹 **VIDEO:** Intro to the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve their Lesson 1 ferry diagram (a rough sketch of the ferry drifting south of Mbita). They now annotate and upgrade this diagram using today's learning. Specific required additions: (1) Label Kisumu port as the origin O(0, 0). (2) Write the column vector for the engine thrust (20, 0) next to the eastward arrow. (3) Write the column vector for the lake current (0, −8) next to the southward arrow. (4) Mark and label Mbita's position vector as OA = (60, 0) using a suitable scale. (5) Mark the ferry's actual landing point B and write its position vector OB, leaving the bottom entry blank with a question mark — acknowledging that they do not yet know the combined result precisely. (6) In a box at the bottom of the diagram, learners write: 'To find OB exactly, I will need to combine the two column vectors. This is the next step.' Completed annotated diagrams are held in learners' portfolios as a growing cumulative model of the phenomenon.	1. Say: 'Take out your Lesson 1 ferry diagram. We are going to upgrade it using today's tools. Do not redraw it — annotate it. Add the six items I am about to show you.' 2. Project or write the six annotation requirements one at a time. After each one, allow 1 minute for learners to add it before moving to the next. 3. After item 3 (column vectors for engine and current), pause and say: 'Hold up your diagram. I should be able to see two column vectors written next to two arrows. Show me.' Scan the room visually. Redirect any learner who has written scalars rather than column vectors. 4. After item 5 (position vector of B with a question mark), say: 'Why is the bottom entry of OB still a question mark? Turn and tell your partner — 30 seconds.' Cold-call 1 pair. Affirm: 'We know the ferry travelled 60 km east, so the top entry is 60. But we do not yet have the tool to calculate how far south it ended up. That tool is vector addition — coming in Lesson 3.' 5. After item 6 (the box statement), say: 'Read your box	Cumulative Model Annotation — rather than drawing a new diagram each lesson, learners build on a single evolving model of the ferry phenomenon. This strategy makes conceptual growth visible and tangible. The deliberate use of a question mark in OB's bottom entry is a pedagogical move that names the productive gap — learners leave the lesson knowing exactly what they do not yet know, which motivates Lesson 3. This mirrors authentic scientific modelling, where models are always works-in-progress.	Observable indicators: (1) Learner's annotated diagram shows two correctly written column vectors with appropriate signs — (20, 0) for engine, (0, −8) for current. (2) Mbita is correctly labelled with position vector (60, 0) at the right scale. (3) The question mark entry for OB is present and the learner's box statement correctly identifies vector addition as the missing tool. Red flag: a learner who fills in OB as (60, −8) without explanation has intuited vector addition but has not yet been taught to justify it — celebrate this intuition publicly and ask the learner to 'hold that thought for Lesson 3, because you may be exactly right.'

		statement aloud to your partner.' Allow 30 seconds. 6. Collect or photograph 4–5 annotated diagrams to display as anchor models for Lesson 3.		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) When learners completed the column vector table in Task A, what were the most common sign errors — did more learners make errors on horizontal negatives (leftward arrows) or vertical negatives (downward arrows)? Plan a 5-minute warm-up correction for Lesson 3 if more than 30% of the class made sign errors. (2) Look at the yellow sticky notes on the DQB: are most learners asking about combining vectors (which is the intended next step) or are they asking about something else entirely? If learners are asking about magnitude or direction rather than addition, consider whether the bridge to Lesson 3 needs re-sequencing. (3) In the model-building phase, how many learners spontaneously wrote (60, −?) or (60, −8) for OB before being told to use a question mark? These learners are reasoning ahead of instruction — consider giving them an extension task at the start of Lesson 3 to formalise their intuition. (4) Did the Nairobi Grid paired task generate genuine mathematical talk between partners, or did one partner dominate? If collaboration was uneven, restructure the Lesson 3 pair activity using assigned roles (recorder and explainer, alternating per question).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that any directed arrow on a grid can be fully described by counting its horizontal steps (right positive, left negative) and vertical steps (up positive, down negative), and recording these as a column vector. They observed that the ferry's engine thrust produces a column vector of (20, 0) — eastward only — while the lake current produces (0, −8) — southward only — and that these two separate column vectors sit inside the same ferry diagram.
What did I learn?	Learners learned that a column vector is a precise mathematical notation that captures both the magnitude and direction of a vector using two signed components. They learned that a position vector describes the location of a point relative to an origin, written in column vector form. They connected these ideas to the ferry scenario by writing the position vector of Mbita as (60, 0) and recognising that the ferry's actual landing point has a non-zero negative bottom entry — a direct mathematical consequence of the southward current.
How does this explain the phenomenon?	Learners explained why the captain's intended path can be written as (60, 0) — with no vertical component — while the actual journey must include a negative vertical component due to the current. They articulated that column vector notation is the mathematical language needed to describe the ferry's drift precisely, and that combining the engine vector and the current vector into one resultant vector is the next mathematical step — which they identified as vector addition.

LESSON 3 (80 minutes): Vector Addition — Triangle and Parallelogram Rules	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to add vectors using the triangle law and parallelogram law, and to apply these methods to find the resultant vector in real-world contexts such as navigation on Lake Victoria.
Knowledge	Learners will know that: (1) two or more vectors can be combined into a single resultant vector; (2) the triangle law states that if two vectors are represented as two sides of a triangle taken in order, the resultant is the closing third side taken in the opposite direction; (3) the parallelogram law states that if two vectors are represented as two adjacent sides of a parallelogram, the resultant is the diagonal from the common point; (4) vector addition is commutative (a + b = b + a) and associative.
Skills	Learners will be able to: (1) draw accurate vector diagrams using the triangle and parallelogram rules; (2) calculate the magnitude of a resultant vector using the Pythagorean theorem and trigonometry; (3) determine the direction of a resultant vector using trigonometric ratios; (4) apply vector addition to solve navigation and displacement problems set in Kenyan contexts.
Attitudes	Learners will appreciate the precision and power of mathematical tools for solving real-life problems, develop persistence when working through multi-step vector problems, and value collaborative reasoning when constructing and critiquing vector diagrams.
Key Inquiry Question	How can vectors be represented, combined, and applied to solve problems involving displacement, navigation, and geometry?
Purpose in Storyline	In Lessons 1 and 2, learners established what vectors are, how to represent them, and how to distinguish them from scalars. They know the ferry experienced two simultaneous directed forces — engine thrust east and current south. Lesson 3 now gives learners the core mathematical machinery to combine those two vectors into a single resultant, directly answering: how far did the ferry actually travel, and in what true direction? This is the turning-point lesson — learners move from describing the problem to solving it.
Safety Notes	No laboratory hazards in this lesson. Ensure learners use rulers and protractors carefully. Remind learners to keep drawing instruments flat on the desk to avoid injury. Supervise use of compasses (drawing compasses) when constructing accurate diagrams.

B. LESSON OVERVIEW
Learners revisit the Kisumu–Mbita ferry scenario and formally combine the eastward engine thrust (20 km/h) and southward current (8 km/h) into a single resultant vector using both the triangle law and the parallelogram law. Through guided drawing activities, peer construction tasks, and a structured sensemaking discussion, learners calculate the ferry's true displacement magnitude (~21.5 km/h resultant speed) and direction (~21.8° south of east) using Pythagoras and trigonometry. The lesson anchors both geometric methods to the same real-world outcome, showing they are equivalent, and extends to a second Kenyan navigation context (a fishing boat near Homa Bay) to consolidate the skill. Learners update their Driving Question Board and revise their cumulative vector model of the ferry's journey.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Area of a	Learners are shown a large projected image of the Lake Victoria ferry scenario from	Display the ferry diagram on the board with the two component vectors drawn (east arrow labeled	Prediction-first activation (Elicit-Before-Explain): By committing to a drawn prediction before	Observable indicator: Do learners draw the resultant arrow as diagonal (south-east direction)? A

[parallelogram on the coordinate plane](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Vector Addition (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Lessons 1 and 2. They are told: 'The ferry engine pushes east at 20 km/h. The current pushes south at 8 km/h. These are two separate vector arrows. Today, before we do any mathematics, draw in your exercise books what you think the ferry's ACTUAL path looks like as a single arrow — its direction and roughly how long it is compared to the two separate arrows.' Learners work individually for 3 minutes, sketching their predicted resultant arrow. They then share predictions with a shoulder partner (Think-Pair) for 2 minutes, noting agreements and disagreements.	20 km/h, south arrow labeled 8 km/h) but NO resultant arrow. Say verbatim: 'Look at these two arrows. They are both acting on the ferry AT THE SAME TIME. In your book, draw what you think the ferry's single true journey arrow looks like — its direction and its size relative to these two arrows. You have 3 minutes. Start now.' Set a visible 3-minute timer. Circulate silently for the first 90 seconds — do NOT answer questions; allow productive struggle. After 3 minutes, say: 'Now turn to your partner — compare your arrows. Do they point the same direction? Is one longer than the other? You have 2 minutes.' After pair discussion, cold-call 3 students (choose one who drew a longer arrow, one who drew a shorter arrow, one who is uncertain) to share. Write their three predictions as rough sketches on the board under the heading 'Our Predictions.' Say: 'We will come back to these predictions at the end of Phase 2 and see who was closest — and why.'	instruction, learners activate prior intuitions about direction and magnitude, creating a cognitive hook. Disagreements between partners surface the core misconception that the resultant is simply 'between' the two vectors in length, setting up the need for the precise geometric rules to follow.	correct directional intuition is a green flag. Red flags: learners who draw the resultant pointing due east (ignoring the current), due south (ignoring the engine), or who draw two separate arrows rather than one combined arrow. Teacher notes these on a quick mental tally during circulation — these learners are targeted for closer support in Phase 2.
Observe Phase 📹 **VIDEO:** [Area of a parallelogram on the coordinate plane](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Vector Addition (Flexbook)](#) Source: CK-12	Learners engage in a structured guided-drawing investigation in two parts. PART A — Triangle Law (30 min): Each learner receives a printed 1 cm : 2 km/h grid sheet (or draws a grid in their books). Step-by-step, guided by the teacher, they: (1) draw vector a (east, 10 cm representing 20 km/h) from a starting point O; (2) from the tip of a, draw vector b (south, 4 cm representing 8 km/h); (3) draw the closing side from O to the tip of b and label it 'Resultant R.' They measure R with a ruler and record its length in cm, then convert back	Distribute grid sheets. Say verbatim: 'We are going to draw the ferry's journey mathematically. Every centimetre on your paper equals 2 km/h. The engine pushes east at 20 km/h — how many centimetres is that?' Wait 10 seconds. Cold-call 2 students. Confirm: '10 centimetres — good.' Demonstrate Step 1 on the board with a large ruler. 'Draw your east arrow now — 10 cm, pointing right, starting at point O. Label it a = 20 km/h east.' Circulate and check all learners have a correct horizontal arrow before proceeding. Then	Guided Inquiry Drawing with Iterative Verification: Learners construct both geometric methods independently on paper, then directly compare outcomes. This 'twin-method convergence' strategy is powerful because when learners see that both methods produce the same resultant arrow, they develop confidence that the mathematics is internally consistent and not arbitrary. The physical act of drawing, measuring, and converting reinforces procedural fluency while the comparison builds conceptual	Observable indicators: (1) Learners correctly place the second vector at the TIP of the first (not back at O) for the triangle law — a common error is starting both vectors at O; teacher scans for this during circulation and corrects immediately. (2) Measured resultant length should be approximately 10.77 cm (±0.3 cm acceptable); angle should be approximately 21–22° south of east. (3) Parallelogram diagonal should match triangle resultant within measurement error. Teacher uses a quick show-of-

🔍 Search ARES for similar readings	to km/h. They use a protractor to measure the angle R makes with the east direction. PART B — Parallelogram Law (15 min): Starting from the same point O, learners draw both vectors as adjacent sides of a parallelogram (a east, b south), complete the parallelogram, and draw the diagonal from O. They measure the diagonal, compare it to their triangle-law resultant, and record: 'Are they the same?'	say: 'The current pushes south at 8 km/h. South on our diagram is downward. From the very TIP of your east arrow — not from O, from the tip — draw a downward arrow. How many centimetres?' Wait 10 seconds. Cold-call 2 students. Confirm: '4 centimetres.' Demonstrate. 'Draw it now. Label it b = 8 km/h south.' Circulate again. Say: 'Now draw a straight line from O — your starting point — to the tip of that south arrow. That line is the resultant. Label it R.' Pause 15 seconds for drawing. 'Measure R in centimetres with your ruler. Write it down. Convert it: multiply by 2 to get km/h.' After measurement, say: 'Now use your protractor. Measure the angle between R and the east direction. Write it down.' Cold-call 3 students for their measurements. Write values on board. Say: 'Interesting — they are all close to the same angle. Why might there be small differences?' Accept 2 responses. Transition: 'Now we try a second method — the parallelogram law. Watch carefully.' Demonstrate the parallelogram construction. Learners construct independently. After both parts, say: 'Compare your two resultants — triangle method and parallelogram method. Are they the same length and direction?' Cold-call 4 students. Confirm equivalence.	understanding.	hands: 'Raise your hand if your two methods gave you the same resultant.' If fewer than 70% raise hands, pause and reteach the parallelogram construction before proceeding.
Explain Phase 📹 **VIDEO:** Area of a parallelogram on the coordinate plane Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners move from drawing to calculation. Using the values from Phase 2, the class works through the algebraic verification: $\|R\| = \sqrt{(20^2 + 8^2)} = \sqrt{(400 + 64)} = \sqrt{464} \approx 21.5$ km/h, and direction $\theta = \arctan(8/20) \approx 21.8°$ south of east. Learners then tackle a parallel Kenyan context: 'A fishing boat	Write the ferry vectors on the board: 'a = 20 km/h east, b = 8 km/h south.' Say verbatim: 'Your ruler gave us approximately 21.5 km/h. Now let us PROVE it with algebra. These two vectors are perpendicular — the engine is east, the current is south — so what theorem can we use to find	Connect-and-Extend with Parallel Context: The algebraic calculation of the ferry resultant directly validates the drawn measurement (connecting), and the Homa Bay problem extends the same skill to a new orientation (north-east instead of east-south), forcing learners to generalise the method	Observable indicators: (1) Learners correctly apply Pythagoras to perpendicular vectors without prompting. (2) Learners correctly identify which sides are 'opposite' and 'adjacent' for the arctan calculation relative to the angle they are finding. (3) Homa Bay answer: $\|R\| = \sqrt{(15^2 +$

for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	near Homa Bay heads due north at 15 km/h to reach a landing point. A wind blows it due east at 6 km/h. Draw the vector triangle, calculate the resultant speed, and find the angle east of north the boat actually travels.' Learners work in groups of four, assigning roles: Drawer, Calculator, Checker, Reporter. Groups present solutions to the class.	the resultant?' Wait 15 seconds. Cold-call 2 students. Guide toward Pythagoras. Write: '$\|R\|^2 = 20^2 + 8^2$.' Say: 'Calculate this with me.' Pause 10 seconds for working. Cold-call 1 student for the answer. Write $\sqrt{464} \approx 21.5$ on board. Say: 'So the ferry's true speed is about 21.5 km/h. Now the direction — the angle south of east. Which trigonometric ratio connects the opposite side and the adjacent side?' Wait 10 seconds. Cold-call 2 students — confirm 'tangent.' Write: '$\tan \theta = 8/20 = 0.4$, so $\theta = \arctan(0.4) \approx 21.8°$.' Say: 'So the ferry travelled at 21.8 degrees south of east. Does anyone remember our predictions from Phase 1? Let us look at the board.' Return to predictions and compare. Then say: 'Now your turn — Homa Bay fishing boat. I want you in groups of four. Drawer draws the triangle. Calculator does the Pythagoras and arctan. Checker verifies every step. Reporter presents. You have 10 minutes.' Circulate actively — spend at least 60 seconds with each group. Listen for whether groups correctly identify north and east as perpendicular. After 10 minutes, cold-call 2 groups to present. After each, ask the class: 'Does the Reporter's diagram match their calculation? Thumbs up or thumbs sideways.' Address any discrepancies.	rather than memorise a specific case. Group roles (Drawer, Calculator, Checker, Reporter) distribute cognitive load and create accountable talk, where every group member must understand the solution well enough to spot errors.	$6^2) = \sqrt{(225 + 36)} = \sqrt{261} \approx 16.2$ km/h, direction ≈ 21.8° east of north — teacher checks group whiteboards or books for these values. (4) Listen for the phrase 'perpendicular' in group discussions — this signals learners understand WHY Pythagoras applies; absence of this word is a flag for misconception.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Area of a parallelogram on the coordinate	Learners individually write a 'What We Now Know' sticky note (or entry in their DQB section of their books) responding to the prompt: 'Using today's lesson, what can you now say about WHY the ferry missed Mbita and HOW FAR off-course it was?' They also write one	Display the class Driving Question Board (a large poster or section of the board that has been accumulating across all lessons). Say verbatim: 'Look at the big question: Why did the Kisumu ferry miss Mbita? Based on today, you now have real mathematical	Driving Question Board as a Running Evidence Ledger: The DQB makes the progression of understanding visible over the entire unit. Placing today's evidence explicitly under 'Evidence We Have' gives learners a sense of epistemic progress — they are	Observable indicators: (1) 'Know' notes should reference the resultant vector, approximate magnitude (~21.5 km/h or ~21.8°), and the idea that two perpendicular vectors combined. (2) Vague notes ('the current moved the ferry') without

[plane](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Vector Addition (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	remaining question: 'What do we still need to figure out to fully solve the ferry problem or to help the captain plan better?' Sticky notes are placed on the class DQB under two columns: 'Evidence We Have' and 'Questions We Still Have.' Teacher reads out 4–5 notes aloud.	evidence. Take 3 minutes — write one thing you now KNOW about why the ferry drifted and how far, and one question you still have.' Set a 3-minute timer. Circulate silently. After 3 minutes, collect/post notes. Read 3 'Know' notes aloud — after each, say: 'Class, do you agree? Thumbs up.' Read 2 'Still wondering' notes and say: 'Excellent — hold these questions. Some will be answered in Lesson 4 when we look at resolving vectors and planning a corrected course.' Point to the DQB headline question and say: 'We are getting closer. The ferry missed because two vectors combined into a resultant that was NOT due east. We can now calculate exactly how far south it landed. What we cannot yet do is tell the captain how to AIM the engine differently to cancel the current — that is our next challenge.'	not just learning isolated skills but building a case. The 'Questions We Still Have' column creates forward momentum, making Lesson 4 feel necessary and anticipated rather than arbitrarily assigned.	mathematical specificity indicate the learner has conceptual understanding but lacks procedural consolidation — flag for follow-up. (3) 'Still wondering' notes that ask about correcting course, working backwards, or three vectors indicate high readiness for extension work in Lesson 4.
Model Building Phase 📹 **VIDEO:** [Area of a parallelogram on the coordinate plane](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **HTML:** [Vector Addition (Flexbook)](#) Source: CK-12 🔍 [Search ARES for similar readings](#)	Learners return to the cumulative ferry model they have been building since Lesson 1 (a diagram in their books or on a shared class poster). Today they add: (1) the triangle-law construction on their diagram — the two component vectors (east engine, south current) drawn tip-to-tail with the resultant R closing the triangle; (2) labels: a = 20 km/h east, b = 8 km/h south, R ≈ 21.5 km/h at 21.8° S of E; (3) a written annotation: 'The triangle/parallelogram rule shows the actual path the ferry took.' Learners also add a small 'scale bar' to their diagram. Pairs compare models and add any missing labels before the lesson closes.	Say verbatim: 'Open your ferry model from Lesson 1. We are going to make it more mathematically complete today. You need to add three things: the two component vectors drawn tip-to-tail, the resultant R with its magnitude and angle, and a written sentence explaining what the triangle rule shows. You have 5 minutes.' Set timer. Circulate and observe: check that resultant arrow is longer than the south component (common error: learners draw R shorter than b); check that the angle is measured from the east direction, not from the south component. After 5 minutes, ask pairs to compare and add anything missing. Cold-call 2 students to describe their updated	Cumulative Model Revision (Progressive Diagrammatic Modelling): Each lesson adds a mathematically richer layer to the same base diagram of the ferry scenario. This strategy externalises conceptual growth — learners can literally see, by looking at their model, how much more they understand than in Lesson 1. Annotating with both numerical values and a written sentence forces integration of procedural results and conceptual meaning, addressing the common gap where learners can calculate but cannot explain.	Observable indicators: (1) The resultant arrow R is drawn from the origin O to the final tip of the second component vector (correct triangle law). (2) The magnitude label reads approximately 21.5 km/h and the angle approximately 21.8° south of east. (3) The written sentence uses the words 'resultant,' 'triangle rule' or 'parallelogram rule,' and references the ferry context — purely abstract sentences ('vectors add to give a resultant') without context reference indicate surface-level understanding. (4) Scale bar is present and consistent with the vector lengths drawn.

		model to the class in one sentence each. Close by saying: 'Your model now shows the full mathematical story of why the ferry drifted. In Lesson 4, we will add the next chapter: how could the captain have aimed differently to arrive exactly at Mbita? That will require us to work backwards — resolving vectors into components. Your model will be ready for that upgrade.'		

D. TEACHER REFLECTION
After the lesson, reflect on the following questions: (1) DRAWING ACCURACY — Did the majority of learners successfully construct both the triangle-law and parallelogram-law diagrams with the correct orientation? If many learners placed the second vector back at O instead of at the tip of the first, plan a 5-minute warm-up in Lesson 4 where learners practise tip-to-tail placement with simple 1D examples before extending to 2D. (2) PYTHAGORAS TRANSFER — Were learners able to independently invoke Pythagoras for the resultant magnitude, or did they need significant prompting? If more than 30% needed prompting, consider including a brief 'right triangle review' card in Lesson 4's materials. (3) DIRECTION CALCULATION — Did learners correctly identify which angle to find and which sides to use for arctan? A common error is using $\tan \theta = 20/8$ instead of $8/20$, giving the complementary angle. Note which learners made this error for targeted support. (4) DQB QUALITY — Were the 'Still wondering' questions pointing forward toward course correction and vector resolution (ideal), or were they still asking questions already answered in Lessons 1–2 (indicating the storyline is not yet coherent for those learners)? Adjust the Lesson 4 introduction accordingly. (5) ENGAGEMENT — The Homa Bay fishing boat context was introduced to broaden Kenyan relevance beyond Kisumu. Note whether learners engaged more readily with this second context or found it disorienting — this informs whether future lessons should use a single running context or multiple parallel Kenyan contexts.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed two simultaneous vector diagrams of the Kisumu–Mbita ferry — an eastward engine thrust (20 km/h) and a southward current (8 km/h) — and constructed both the triangle-law and parallelogram-law diagrams on grid paper, measuring the resultant arrow's length and direction with rulers and protractors.
What did I learn?	Learners learned that two vectors can be combined into a single resultant vector using the triangle law (tip-to-tail) or the parallelogram law (diagonal of adjacent sides), that both methods yield the same resultant, that the magnitude can be calculated using the Pythagorean theorem ($\|R\| = \sqrt{20^2 + 8^2} \approx 21.5$ km/h), and that the direction is found using arctan ($\theta = \arctan(8/20) \approx 21.8°$ south of east).
How does this explain the phenomenon?	Learners explained that the Kisumu ferry missed Mbita because the southward current vector combined with the eastward engine vector to produce a resultant displacement directed approximately 21.8° south of east — the ferry did not travel due east but along the resultant path, landing south of the intended destination by a calculable amount.

LESSON 4 (40 minutes): Subtraction of Vectors and the Zero Vector — Charting the Correction Course Back to Mbita	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners will understand vector subtraction as the operation that allows a navigator — or any problem-solver — to find the correction needed to get from an unintended position back to a desired destination.
Knowledge	Learners will be able to define the zero vector and the negative of a vector; state that subtracting vector b from vector a is equivalent to adding the negative of b to a; perform vector subtraction algebraically using column vector notation and geometrically using directed line segments.
Skills	Learners will subtract vectors algebraically using column vectors; represent vector subtraction geometrically on a coordinate grid; identify the zero vector as the result when a vector is added to its negative; construct a correction vector that closes a vector path back to a starting or intended point.
Attitudes	Learners will appreciate that mathematical precision in direction matters in real life — an error of a few degrees or kilometres can strand hundreds of people — and will develop persistence in checking vector calculations for both magnitude and direction.
Key Inquiry Question	How can vectors be represented, combined, and applied to solve problems involving displacement, navigation, and geometry?
Purpose in Storyline	In Lesson 3 learners discovered where the ferry actually landed by adding the engine vector and current vector. Now the question shifts: what correction vector must the captain radio to the crew to get the vessel from its wrong landing point back to Mbita? This is a subtraction problem. Vector subtraction is the natural next tool in the growing correction story, and it advances the Driving Question Board by giving learners a concrete mathematical means of undoing unwanted displacement — a critical step toward fully solving the ferry problem in Lesson 8.
Safety Notes	No physical hazards. Remind learners to use rulers carefully when drawing directed line segments. Ensure grid paper is shared equitably across groups.

B. LESSON OVERVIEW

The lesson opens by reminding learners that the ferry landed south of Mbita after combining the engine vector and the lake current vector. A short prediction task asks learners: if you know where you are and where you need to be, what single vector describes the journey back? Learners observe a geometric construction on the board showing the resultant from Lesson 3 and the intended destination, then reason that the missing vector is the difference between the two. The Explain phase formalises the negative of a vector, the zero vector, and subtraction as addition of the negative, with column-vector algebra and a grid-based diagram. A supporting context — a Rift Valley farmer pulling a rope cart that overshoots a storage shed and must be guided back — reinforces the concept in a second Kenyan setting. Learners then build an updated model showing the full three-vector closed path: engine vector, current vector, and correction vector, confirming the zero vector result. The DQB is updated with the new insight.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners are shown a large	Display the carry-over grid from	Anchoring back to the	Teacher circulates during

📹 **VIDEO:** [Significance test for a proportion free response (part 2 with correction)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	laminated grid on the board (or drawn on the chalkboard) that carries over the result from Lesson 3: point K (Kisumu) at the origin, point M (Mbita) at (60, 0), and point L (actual landing point) at (60, -16) — representing the ferry drifting 16 km south after 2 hours on a 20 km/h eastward engine with an 8 km/h southward current. The teacher points to L and M and poses the Predict Prompt in writing on the board: 'The ferry is now stuck at L. The captain needs to reach M. Without any calculations yet, draw on your grid paper ONE arrow that you think shows the journey from L to M. Then write one sentence describing what that arrow represents.' Learners work individually for 3 minutes, sketching the arrow and writing their sentence. Pairs then share predictions verbally for 1 minute. The teacher records four or five predicted descriptions on the board verbatim without correcting them, e.g. 'go back north', 'reverse the drift', 'a new push vector', 'the opposite of the current'.	Lesson 3 prominently. Read the predict prompt aloud and say: 'Think quietly first. Do not call out. I want every hand drawing before any mouth speaks.' Allow 3 minutes of silent individual work. Then say: 'Turn to the person next to you. Share your arrow direction and your sentence. You have 1 minute.' After pairs share, cold-call three or four pairs and record their language precisely on a side section of the board labeled PREDICTIONS. Do not affirm or correct. Say: 'We are going to test each of these ideas. Hold them in your mind.'	phenomenon — the stranded ferry — gives immediate purpose. Recording learner language without correction preserves cognitive tension and gives the teacher diagnostic data about whether learners are thinking about the vector from L to M or incorrectly from M to L.	individual drawing time and notes: Does the learner draw the arrow from L pointing toward M (north, since M is directly above L on the grid)? Does any learner draw the arrow in the wrong direction (e.g. from M to L)? This reveals whether learners understand that vector direction is defined by start and end point — a key prerequisite for subtraction. Teacher makes a mental note of any learner who draws the correction arrow correctly and labels it for strategic use in the Explain phase.
Observe Phase 📹 **VIDEO:** [Significance test for a proportion free response (part 2 with correction)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12	The teacher constructs a second diagram live on the board — or reveals a pre-drawn version on chart paper — that shows the following: (1) The engine vector e drawn from K(0,0) to (60, 0) labeled as e = (60 above 0). (2) The current vector c drawn tip-to-tail from (60, 0) to L(60, -16), labeled c = (0 above -16). (3) The resultant r drawn from K to L, labeled r = e + c = (60 above -16). (4) A dashed arrow drawn from L(60, -16) to M(60, 0), labeled with a question mark. The teacher says: 'Look at the four vectors in this diagram. Three are labelled.	Construct or reveal the diagram with clear labels and grid coordinates. Point to the dashed arrow and say: 'I am not telling you what this vector is yet. You are going to figure it out from what you can see. Write your observations — not guesses, observations.' Give 4 minutes of uninterrupted writing time. Then say: 'Now look at the Nakuru cart problem. Sketch it on your grid. What are the coordinates of the correction arrow?' Give 2 minutes. WAIT TIME: After asking 'What are the horizontal and vertical components of the mystery vector?', wait a full	The two-scenario observe phase (ferry and rope cart) uses a second Kenyan context to make the concept of a correction vector feel broadly applicable, not specific to one situation. The deliberate withholding of the label on the mystery vector forces learners to reason from the diagram rather than memorise a definition. The grid-based observation connects the geometric representation to the algebraic one that follows.	Teacher asks two or three learners to read one observation each. Listen for: Does the learner identify the start point as L and end point as M? Does the learner state the components as (0 above 16) — i.e., 16 units north? Does any learner spontaneously name this as M minus L or r minus c? These responses indicate readiness for algebraic formalisation. If most learners identify only the direction but not the components, the teacher knows more scaffolding is needed in the Explain phase before moving to algebra.

🔍 Search ARES for similar readings	One is not. Observe carefully: where does the mystery vector start? Where does it end? What are its horizontal and vertical components?' Learners write three observations in their notebooks under the heading WHAT I NOTICE. They are given 4 minutes for this. Learners are then asked to observe a second supporting scenario displayed on a small poster or written on the board: 'A farmer in Nakuru uses a rope to pull a cart from a shed at point P(2, 3) to a maize store at Q(7, 3). The rope slips and the cart ends up at R(7, 6) instead. What vector moves the cart from R back to Q?' Learners sketch this on their grid paper.	8 seconds before accepting responses. This wait time is critical — it allows learners who process spatially to catch up with those who process symbolically.		
Explain Phase 📹 **VIDEO:** Significance test for a proportion free response (part 2 with correction) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces formal vocabulary and notation in three clearly signposted steps, each with a brief learner activity embedded. STEP 1 — THE NEGATIVE VECTOR: Teacher writes on the board: 'If vector a = (3 above 5), then the negative of a, written -a, equals (-3 above -5). The negative vector has the same magnitude but the opposite direction.' Learners write this in their notebooks, then answer: 'If the current vector c = (0 above -16), write -c.' (Answer: (0 above 16).) Learners draw both c and -c on a small grid, noting they point in opposite directions along the same line. STEP 2 — THE ZERO VECTOR: Teacher writes: 'When you add a vector to its negative, you get the zero vector: a + (-a) = 0, where 0 = (0 above 0). The zero vector has no magnitude and no direction. It means you are back where you started.' Teacher connects this to the ferry: 'If the	Use three large numbered headings — STEP 1, STEP 2, STEP 3 — so learners track the logical progression. After each step, pause and ask: 'Before I continue, write the column vector answer in your notebook. No talking.' Give 60 seconds of silent individual work. Then say: 'Check with your neighbour.' After Step 3, cold-call a learner to explain in words — not numbers — what the correction vector means for the captain. Specific quote: 'Achieng, without pointing at the board, tell me in your own words what the correction vector tells the captain to do.' WAIT TIME: After posing this question, wait 10 seconds. Do not rescue Achieng with hints. If she is silent after 10 seconds, say: 'Take five more seconds. You can do this.' If still silent, say: 'Tell me one thing you know about where the ferry is.' This scaffolding preserves dignity while maintaining rigour. After learners complete the	The three-step progression (negative vector, zero vector, subtraction) mirrors the logical dependency: you cannot define subtraction without the negative, and you cannot make sense of the negative without the zero vector as a reference point. Embedding a short learner-practice moment after each step prevents passive listening. Using position vectors for M and L to derive the correction grounds the abstract subtraction in a concrete geometric act — the ferry moving from one labelled point to another on the grid. The Nakuru cart problem acts as a transfer check within the Explain phase.	Exit check at the end of the Explain phase: Teacher writes on the board: 'Vector p = (5 above -3) and vector q = (2 above 4). Find p - q.' Learners write the answer on a small piece of paper or in their notebooks and hold it up or the teacher circulates to read answers in 90 seconds. Correct answer: (3 above -7). This takes 2 minutes total and tells the teacher immediately whether the class is ready to model-build or needs a further worked example. Teacher records names of learners who write incorrect answers for targeted support during Model Building.

	ferry travels the resultant r from K to L, then travels -r from L back to K, the total displacement is zero.' Learners write the column-vector proof: r + (-r) = (60 above -16) + (-60 above 16) = (0 above 0). STEP 3 — VECTOR SUBTRACTION: Teacher writes: 'Vector subtraction: a - b means a + (-b). In column vectors: (p above q) - (r above s) = (p-r above q-s).' Teacher models the ferry correction: 'The correction vector from L to M equals the position of M minus the position of L. Let position vector of M = m = (60 above 0) and position of L = l = (60 above -16). Correction = m - l = (60 above 0) - (60 above -16) = (60-60 above 0-(-16)) = (0 above 16).' Teacher says: 'This matches the dashed arrow we observed — 16 units due north. The ferry must travel 16 km due north to reach Mbita from its wrong landing point.' Learners individually complete the Nakuru cart problem using the same method: correction from R(7,6) to Q(7,3) = q - r = (7 above 3) - (7 above 6) = (0 above -3). Two learners write solutions on the board while the class checks.	cart problem on the board, ask the class: 'Does the group on the board agree with your answer? If not, who is correct and why?' This promotes mathematical argumentation.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Significance test for a proportion free response (part 2 with correction)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Groups return to the class Driving Question Board — a large chart or section of the chalkboard divided into columns: WHAT WE KNOW, WHAT WE THINK, WHAT WE STILL WONDER, EVIDENCE. Each group has been maintaining a small group DQB copy in their exercise books. The teacher poses two specific DQB update prompts in writing: (1) 'Write one new thing we now KNOW about the ferry journey that we did not know after Lesson 3.' (2) 'Write one question the correction vector raises that we	Keep this phase to no more than 5 minutes. Use a timer visible to the class. Say: 'Groups have 60 seconds to write their two entries. Efficiency matters — this is not an essay, it is a precise claim or a precise question.' After postings, point to the EVIDENCE column and ask: 'What mathematical evidence from today supports what you wrote in KNOW? Kipkorir, read your group KNOW entry and tell me the column-vector calculation that proves it.' This move explicitly links DQB knowledge claims to	The DQB update consolidates new knowledge in learners own language and simultaneously surfaces productive questions for future lessons. Requiring evidence for KNOW entries prevents the DQB from becoming a list of unsubstantiated claims. The star-marking of a selected WONDER entry creates narrative continuity across lessons and signals to learners that their questions are taken seriously and will be answered.	Teacher reads all posted KNOW entries before closing the DQB phase. If any group writes a KNOW entry that contains a misconception — e.g., 'the correction vector cancels the whole journey' (confusing correction from L to M with the zero vector) — the teacher does not correct it publicly but instead says: 'Interesting claim. What evidence supports it? Can you write the column vector that proves it?' This deflects the correction back to the learner and creates a

📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	cannot yet answer.' Groups write their entries on sticky notes or torn paper slips (60 seconds), then a representative posts them under the correct DQB column. The teacher reads two or three entries aloud. Anticipated new KNOW entry: 'The ferry needs a correction vector of (0 above 16) — 16 km due north — to reach Mbita from where it landed.' Anticipated new WONDER entry: 'Can the captain apply the correction while the ferry is moving, or does it have to stop first?' or 'What if the wind changes direction during the correction?' The teacher highlights one wonder question that will be relevant to Lesson 8 and says: 'Mark this question with a star. We will return to it in our final lesson when we solve the full problem.'	mathematical evidence, building scientific and mathematical argumentation habit. Specific quote for transitioning out of DQB: 'Our board is growing. Every lesson, the captain gets one more tool. Today the captain learned how to calculate the correction. In Lesson 8, the captain will use all these tools together to never miss Mbita again.'		self-checking moment without public embarrassment.
Model Building Phase 📹 **VIDEO:** [Significance test for a proportion free response (part 2 with correction)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners work in groups of three or four to build a complete cumulative vector model of the ferry journey so far. On A4 grid paper, each group draws: (1) Engine vector e = (60 above 0) from K to point A. (2) Current vector c = (0 above -16) from A to L (actual landing). (3) Correction vector v = m - l = (0 above 16) from L to M. (4) They verify that e + c + v = (60 above 0) + (0 above -16) + (0 above 16) = (60 above 0). They label this final result as the vector from K to M, which equals the intended journey. Groups then answer three structured questions in writing: Q1 — 'What is the column vector of the correction v? Show the subtraction that produces it.' Q2 — 'If a second current of (3 above 0) were added (an eastward push), would the correction vector change? If yes, write the new correction.' Q3 —	Circulate during group work. Do not answer content questions immediately. Instead ask: 'What do you know about vector subtraction that could help here?' or 'What does the zero vector tell you about a round trip?' For Q2 — the extension question — give an additional 30 seconds of wait time before engaging groups that are stuck: say 'Rewrite the new landing point coordinates first, then subtract from M.' For the presenting group, after their explanation say to the class: 'Do you agree with this group? If you got a different answer for Q2, stand up and explain your reasoning.' This creates a productive classroom disagreement moment that deepens understanding. Close the model-building phase by pointing to the complete diagram and saying: 'Every arrow on this	Building the complete three-vector closed path and verifying the algebraic sum gives learners the experience of the zero vector as a meaningful geometric and algebraic object — not an abstraction. Q2 extends thinking by varying a parameter, which promotes flexible application over rote procedure. The public display and class challenge of the presenting group model scientific norm of peer review. The closing teacher statement reconnects the mathematics explicitly to the Driving Question.	Teacher collects the A4 grid paper models at the end of the lesson (or photographs them) and checks three things: (1) Are all three arrows drawn in the correct direction with correct start and end points? (2) Is the subtraction in Q1 shown as column-vector working, not just the answer? (3) Does Q3 show the round-trip equation correctly as e + c + (-e - c) = (0 above 0) or equivalent? Learners who complete Q2 correctly with the new correction vector (0 above 16) minus any eastward drift adjustment have demonstrated transfer beyond the lesson. These learners are noted for extension challenge in Lesson 5.

	'The zero vector appears when a vector is added to its negative. Write the column vector equation that shows the round trip K to L and back to K.' At least one group is invited to display their diagram on the board and explain one of the three questions to the class.	diagram represents a real physical event in the ferry crossing. The captain can now calculate any correction — not guess it. That is the power of vectors.'		

D. TEACHER REFLECTION

After the lesson, consider the following: Did learners consistently distinguish between the direction of the correction vector (from wrong point to intended point) and its negative (from intended point to wrong point)? If confusion persisted, plan a brief five-minute warm-up for Lesson 5 using two or three labelled points on a grid and asking: 'Write the vector from A to B. Now write the vector from B to A. What is their sum?' Was the zero vector understood as a meaningful result — the round trip — rather than just a trivial calculation? If learners treated it as unimportant, the scalar multiplication lesson (Lesson 5) should include a brief revisit of the zero vector as a boundary case when a scalar of zero is applied. Did the DQB update produce genuinely new KNOW entries grounded in column-vector evidence, or did groups write vague narrative statements? If the latter, model a precise KNOW entry format at the start of Lesson 5: 'We know [claim] because [column-vector calculation].' Finally, note which learners struggled with the sign handling in column-vector subtraction — particularly the subtraction of a negative component such as 0 - (-16) = 16. These learners may need a brief integer arithmetic review at the start of the next lesson, which can be embedded as a 2-minute oral warm-up.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	The ferry landed at point L, which is 16 km south of Mbita (point M), after the engine vector and current vector were combined. A dashed arrow on the diagram connected L to M and had unknown components. A Nakuru farmer also overshot a maize store when pulling a rope cart, landing at the wrong point.
What did I learn?	The negative of a vector reverses its direction while keeping the same magnitude. The zero vector results from adding a vector to its negative and represents no net displacement — a round trip. Vector subtraction a - b is the same as a + (-b) and is carried out component by component in column vector notation. The correction vector from L to M equals the column vector (0 above 16), found by subtracting the position vector of L from the position vector of M.
How does this explain the phenomenon?	The captain can calculate exactly what correction vector is needed to guide the ferry from its wrong landing point to Mbita by subtracting position vectors. The mathematics of vector subtraction gives a precise, reproducible tool for correcting any navigation error caused by drift — whether from a lake current, ocean tide, or wind — and ensures the ferry can always reach the right destination.

LESSON 5 (80 minutes): Scalar Multiplication of Vectors and Parallel Vectors	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to multiply vectors by scalars and identify parallel vectors, connecting these operations to real-world navigation contexts such as the Kisumu–Mbita ferry route on Lake Victoria.
Knowledge	Learners will know that: (1) multiplying a vector by a positive scalar changes its magnitude but preserves its direction; (2) multiplying a vector by a negative scalar reverses its direction; (3) two vectors are parallel if one is a scalar multiple of the other; (4) scalar multiplication is foundational to describing repeated or scaled displacements in navigation.
Skills	Learners will be able to: (1) multiply a vector by a scalar and state the resulting magnitude and direction; (2) determine whether two given vectors are parallel by testing the scalar multiple relationship; (3) represent scalar multiples of vectors geometrically on a diagram; (4) apply scalar multiplication to solve simple navigation and displacement problems.
Attitudes	Learners will: (1) appreciate the precision that directed quantities bring to everyday navigation decisions; (2) show curiosity about how scaling a vector models real physical changes such as engine speed adjustments; (3) demonstrate perseverance when working through multi-step vector problems.
Key Inquiry Question	How can vectors be represented, combined, and applied to solve problems involving displacement, navigation, and geometry?
Purpose in Storyline	In Lessons 1–4, learners built the foundational language of vectors — representation, addition by triangle and parallelogram law, and resultant calculation — to begin explaining why the Kisumu ferry drifted south of Mbita. In Lesson 5, learners now ask: what if the captain doubled the engine thrust, or the current was three times as strong? Scalar multiplication gives learners the tool to scale any vector — modelling speed changes, stronger currents, or repeated equal displacements — and parallel vectors explain why the drifted path and the engine's intended path are mathematically related. This lesson equips learners to build a more complete, quantitative correction model for the ferry captain in the lessons ahead.
Safety Notes	No physical hazards. Ensure rulers and set squares are handled carefully during geometric constructions. Learners working in groups should share materials respectfully.

B. LESSON OVERVIEW

In this 80-minute lesson, learners investigate scalar multiplication of vectors and the concept of parallel vectors through the lens of the Kisumu–Mbita ferry phenomenon. Beginning with a provocative prediction about what happens when the ferry's engine thrust is doubled or the lake current is tripled, learners move through geometric construction, worked examples, and collaborative sensemaking to discover that scaling a vector changes its magnitude without altering (or by reversing) its direction, and that parallelism between vectors is captured precisely by the scalar multiple relationship. The lesson uses a phased PREDICT → OBSERVE → EXPLAIN → DQB → MODEL structure, embedding Kenya-relevant contexts (Lake Victoria ferries, Rift Valley road convoys, Kericho tea-estate lorry routes) and specific formative checkpoints throughout. By the end, learners can multiply any vector by a scalar, test two vectors for parallelism, and articulate how these operations help the ferry captain reason about scaled currents and engine adjustments.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Video: Proportion Playground Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are presented with a short scenario displayed on the board: 'The Kisumu ferry's engine currently pushes the vessel with velocity vector v = (20, 0) km/h due east. The lake current vector is c = (0, −8) km/h due south. The captain's engineer now proposes two changes on separate test runs: (A) Double the engine thrust, and (B) Triple the southward current. Before any calculation, predict in your notebook: (1) What happens to the direction of the engine thrust when it is doubled? (2) What happens to the direction of the current when it is tripled? (3) Do you think the doubled engine thrust and the original engine thrust are pointing in the same direction? Write YES or NO and one sentence of justification.' Learners write individual silent predictions for 4 minutes, then share with a shoulder partner for 2 minutes. Selected pairs share with the class.	Display the scenario on the board and read it aloud once. Say: 'This is a prediction — there are no wrong answers right now. I want to know what your mathematical instinct tells you.' Allow 4 minutes of silent individual writing — enforce silence explicitly: 'No talking yet — I need your own thinking first.' After 4 minutes, say: 'Now turn to your shoulder partner. You have 2 minutes — each person shares their prediction and one reason.' After partner talk, cold-call exactly 3 pairs using the class register (rows 2, 5, and 8) and ask: 'What did your pair predict, and why?' Apply 10 seconds of wait time before redirecting if a pair is silent. Do NOT correct predictions yet — record key student ideas on the board in two columns: 'Direction changes' vs 'Direction stays the same'. Close with: 'We are going to find out today whether your instincts match the mathematics — and connect this to what the ferry captain needs to know.'	Strategy: Elicit–Record–Revisit. Teacher records raw student predictions publicly on the board without evaluation. This creates cognitive investment — learners are motivated to see whether their prediction is validated or overturned, and the board record allows explicit revisiting at the Explain phase. It also surfaces the common misconception that scaling a vector in any way must change its direction.	Observable indicator: Every learner has a written prediction with at least one sentence of justification in their notebook before partner talk begins. Teacher circulates during silent writing and notes (without intervening) any learner who writes that doubling a vector changes its direction — these learners are flagged for targeted cold-calling in the Explain phase. Threshold: If more than 60% of the class predicts direction changes when a vector is scaled by a positive scalar, the teacher anticipates spending an additional 5 minutes on geometric construction in the Observe phase.
Observe Phase 📹 **VIDEO:** Video: Proportion Playground Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns	Learners receive a structured worksheet (or open to a prepared page in their exercise books) with the following three evidence-gathering tasks. TASK A — Geometric Construction (8 min): Using a ruler and pencil, learners draw vector v = (20, 0) to a scale of 1 cm : 5 km/h as a horizontal arrow of length 4 cm pointing right. Directly below it, they draw 2v — an arrow in the same direction of length 8 cm — and then −v — an arrow of length 4 cm pointing left. They label each arrow and note its length and direction. TASK B —	Distribute the worksheet or direct learners to set up the tasks. Say: 'You have 20 minutes for three tasks. Tasks A, B, and C are designed to give you evidence — you are gathering data, just like a scientist. Work individually and in silence for Tasks A and B. I will call time.' Circulate actively during Task A — crouch beside learners and ask: 'What does the length of your arrow represent? What does its direction represent?' For Task B, look specifically for learners who multiply only one component — intervene with: 'Read me what	Strategy: Graduated Evidence Collection (Geometric → Algebraic → Relational). Learners first build physical intuition through drawing (Task A), then practice the algebraic operation (Task B), then apply a relational test (Task C). Each task provides a different type of evidence pointing to the same mathematical truth: scalar multiplication scales magnitude and preserves/reverses direction. This multi-representational approach ensures learners who struggle algebraically can anchor understanding in their geometric	Observable indicators: (1) In Task A, learners' arrows for 2v should be exactly twice the length of v and point in the same direction; −v should be the same length as v but point left. Teacher flags any learner whose −v is shorter than v. (2) In Task B, all three components of each result must be scaled — partial scaling (only one component) is a key misconception to address. (3) In Task C, Pair 3 (e and f) should be identified as NOT parallel (3/6 ≠ 4/9), testing whether learners check both components. Any

(PLIX) Source: CK-12 🔍 Search ARES for similar readings	Column Vector Calculation (6 min): Learners are given three vectors in column form: p = (3, 4), q = (−2, 5), r = (6, 8). They calculate 2p, −3q, and (1/2)r by multiplying each component by the scalar, and write the results. TASK C — Parallel Vectors Table (6 min): Learners are given four pairs of vectors and must decide whether each pair is parallel. Pair 1: a = (4, 6), b = (2, 3). Pair 2: c = (5, −10), d = (−1, 2). Pair 3: e = (3, 4), f = (6, 9). Pair 4: g = (−8, 12), h = (2, −3). For each pair they write the scalar k such that one vector equals k times the other, or write 'NOT parallel'. All tasks are completed individually before a 2-minute group comparison.	scalar multiplication means for a column vector — both components or just one?' For Task C, circulate and ask: 'How did you test whether this pair is parallel?' After all three tasks, say: 'You have 2 minutes — compare your answers with the person next to you. If you disagree, write down both answers — do not erase.' Cold-call 2 learners on Task A (ask them to hold up their book and describe their diagram), 2 learners on Task B (read out their column vector answer), and 2 learners on Task C (state their parallelism conclusion and their k value). Apply 12-second wait time before each cold-call response. Correct calculation errors immediately and explicitly.	drawing, and vice versa.	learner marking Pair 3 as parallel is flagged for the Explain phase.
Explain Phase 📹 **VIDEO:** Video: Proportion Playground Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners first return to their Phase 1 predictions on the board and self-assess: 'Was your prediction correct? Write one sentence in your notebook beginning: I predicted… but the evidence shows…' (3 min). The teacher then leads a structured whole-class explanation (12 min) formalising scalar multiplication and parallel vectors using the ferry context. Learners copy the formal definition, two worked examples, and the parallelism test into their notes. Following the teacher explanation, learners work in pairs on two application problems (10 min): PROBLEM 1 — The ferry's engine thrust vector is e = (20, 0) km/h. After the captain increases speed, the new thrust vector is e_2 = (50, 0) km/h. Show that e_2 = ke for some scalar k, and state whether the engine direction has changed. PROBLEM 2 — A Kericho tea-estate lorry travels along a road modelled by vector u	Point to the prediction board and say: 'Look at what you predicted 35 minutes ago. Take 3 minutes — silently — to write your self-assessment sentence.' Cold-call one learner whose prediction was correct and one whose was overturned: 'Tell us what you predicted, what the evidence showed, and what that means.' (12-second wait time each.) Then transition to direct instruction. Write on the board: 'SCALAR MULTIPLICATION OF VECTORS' and state clearly: 'If v is any vector and k is a real number (a scalar), then kv is a vector with magnitude |k| times the magnitude of v. If k > 0, kv points in the same direction as v. If k < 0, kv points in the opposite direction. If k = 0, the result is the zero vector.' Work through Example 1 explicitly: 'Let the ferry current be c = (0, −8). What is 3c?' Write each step: 3c = 3 × (0, −8) = (0, −24). 'The magnitude is now 24 km/h	Strategy: Predict–Reconcile–Formalise. Learners explicitly compare their raw predictions with observed evidence before receiving formal definitions. This reconciliation step — naming what changed in their thinking — deepens conceptual encoding rather than treating the formal rule as disconnected. The ferry phenomenon grounds every formal statement: 'tripling the current' is not an abstract operation but a physically meaningful change the captain must account for.	Observable indicators: (1) Self-assessment sentences must include both the original prediction and a reference to specific evidence — vague sentences ('I was wrong') are insufficient; teacher prompts: 'What specific evidence overturned your prediction?' (2) In Problem 1, learners must calculate k = 2.5 and explicitly state the direction is unchanged. (3) In Problem 2, learners must find k = −3 and explain that the negative sign means the lorries travel in opposite directions along parallel roads. Any pair that finds |k| but ignores the sign is flagged — the sign carries directional meaning that is critical for the ferry correction model.

	= (3, −1). A second lorry on a parallel road has displacement vector w = (−9, 3). Show that the roads are parallel, find the scalar k, and explain what the negative sign tells us about the lorries' relative directions of travel. Pairs write full solutions with explanation sentences.	southward — three times as strong, same direction.' Work through Example 2 on parallel vectors: 'Are a = (6, −4) and b = (−3, 2) parallel? Test: is b = k × a? Try k = −1/2: −1/2 × (6, −4) = (−3, 2) ✓. Yes, parallel, and the negative scalar means they point in opposite directions.' For pair problems, circulate and use: 'Talk me through your test for parallelism — step by step.' After 10 minutes, cold-call one pair per problem to present on the board while the class checks. Correct precisely, affirming correct method before correcting errors.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Video: Proportion Playground Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each learner takes a sticky note (or tears a small piece of paper) and writes one of three types of contributions to the Driving Question Board: (A) ANSWERED — something the class can now answer about the ferry that they could not answer before this lesson; (B) NEW QUESTION — something today's lesson made them wonder about; or (C) CONNECTION — a link between today's scalar multiplication idea and a previous lesson (e.g., vector addition, resultant magnitude). Learners post their notes on the class DQB under the correct column heading. The teacher reads three notes aloud — one from each column — and responds briefly. The class then reads the driving question together: 'Why did the Kisumu ferry miss Mbita — and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind?' and the teacher asks: 'Where are we now on answering this question?'	Distribute sticky notes or direct learners to tear paper. Say: 'You have 4 minutes — choose one type of contribution: ANSWERED, NEW QUESTION, or CONNECTION. One idea only — write it clearly so someone else can read it.' Circulate and prompt quieter learners: 'What is one thing you now understand about the ferry that you did not understand at the start of today?' After posting, read three notes aloud — choose deliberately: one ANSWERED note that correctly references scalar multiplication (e.g., 'We can now model what happens if the current doubles'), one NEW QUESTION that anticipates Lesson 6 content (e.g., 'Can we use vectors to find the exact correction angle the captain needs?'), and one CONNECTION note linking back to resultant vectors. After reading each, say: 'Who agrees this belongs under ANSWERED/NEW QUESTION/CONNECTION — thumbs up or thumbs sideways.' Close by reading the driving	Strategy: Collaborative Knowledge Tracking via the DQB. The Driving Question Board makes the class's growing understanding visible and cumulative. Requiring learners to categorise their contribution (ANSWERED / NEW QUESTION / CONNECTION) forces metacognitive reflection — they must evaluate what they know, what they still wonder, and how ideas link. This builds the habit of treating mathematics as an evolving, connected body of knowledge rather than isolated procedures.	Observable indicators: (1) Every learner posts a contribution — teacher notes any learner who does not post and follows up individually. (2) ANSWERED notes should reference specific mathematical content from today (scalar multiplication, parallel vectors, magnitude scaling) — vague notes ('we learned about vectors') indicate surface-level processing. (3) The finger-scale reading gives the teacher a rapid whole-class gauge of perceived progress toward the driving question; if the class mode is 1–2 fingers, the teacher makes a note to begin Lesson 6 with an explicit progress summary.

		question aloud together and asking: 'On a scale of one to five fingers, how close are we to fully answering this question?' Count fingers publicly and note the class mode. Say: 'We are making progress. Next lesson we go further.'		
Model Building Phase 📹 **VIDEO:** Video: Proportion Playground Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Multiplying Matrices by a Scalar: Making Matrix Patterns (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners open to their cumulative ferry model diagram — the vector diagram of the Kisumu–Mbita crossing they have been building and revising since Lesson 1. Using today's learning, they add two new annotated elements: (1) SCALED CURRENT — they draw a new dashed vector representing a current that is 1.5 times the original southward current (i.e., 1.5 × (0, −8) = (0, −12) km/h) and label it '1.5c', annotating: 'If the current were 1.5 times stronger, the ferry drifts even further south — scalar multiplication models this.' (2) PARALLEL PATH — they draw a second possible ferry engine path parallel to the original eastward thrust but starting from a point 10 km north of Kisumu, label the direction vector as 'e = (20, 0)', and annotate: 'This parallel path has the same direction vector — it is a scalar multiple relationship that confirms parallelism.' Finally, learners write a 'Captain's Advisory' sentence at the bottom of their model: 'Captain's note: If the current doubles to (0, −16) km/h, the ferry will drift twice as far south — approximately _____ km off course after the 60 km crossing.' Learners calculate the blank using their knowledge of scalar multiplication and share with a partner.	Say: 'Open your cumulative ferry model. Today you are going to make it more powerful by adding two new layers that use what we learned.' Read the two tasks aloud and give learners 9 minutes to work — enforce individual work for the first 6 minutes, then allow 3 minutes of partner comparison. Circulate and ask targeted questions: 'Why did you draw the scaled current in the same direction? What would it look like if the scalar were negative?' and 'How do you know your parallel path is truly parallel — what is the mathematical test?' For the Captain's Advisory calculation, prompt: 'The original drift was 24 km south over 3 hours at 8 km/h. If the current doubles, what changes?' After 9 minutes, cold-call one learner to describe their scaled current addition and one to share their Captain's Advisory calculation aloud (12-second wait time). Affirm correct model additions. Close the lesson by saying: 'Your ferry model is becoming a real navigation tool. In our next lesson, we will use what we know about vectors to ask: what direction should the captain actually aim the engine so the ferry lands at Mbita, not south of it?'	Strategy: Cumulative Model Revision with Annotation. Requiring learners to add to — not replace — their existing ferry diagram makes the growth of understanding visible and physical. Annotations force learners to translate mathematical operations (scalar multiplication) into natural-language captions meaningful to the ferry context. The Captain's Advisory sentence bridges abstract calculation and real-world consequence, reinforcing that mathematics is a tool for decision-making, not just symbolic manipulation.	Observable indicators: (1) The scaled current vector (1.5c) must be drawn in the same direction as the original current and visually longer — teacher checks that arrow length is proportional, not arbitrary. (2) The parallel path must start at a different point from the original thrust but carry the same direction vector label — any learner who draws the same arrow displaced but changes its label is prompted: 'Direction vector means the direction and relative size — has that changed?' (3) The Captain's Advisory blank: the correct answer is approximately 24 km south (since at doubled current (0, −16) km/h, over 3 hours the drift = 48 km — teacher accepts 48 km as correct; learners who write 24 km have used the original current rather than the doubled one and need immediate correction).

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During Phase 2 Task C (parallel vectors), did learners check BOTH components of a vector pair, or did they test only the first component? If the majority checked only one component, plan a brief whole-class re-teaching moment at the start of Lesson 6 using a counterexample. (2) In Phase 3 Problem 2 (the Kericho lorry problem), did learners articulate the meaning of the negative scalar (opposite direction of travel) in words, or did they simply find $k = -3$ without interpretation? The directional meaning of the sign is critical for the ferry correction model — if interpretation was weak, embed a 'what does the sign tell us?' prompt into every vector problem in Lesson 6. (3) Were the DQB NEW QUESTION contributions pointing toward the correction angle/bearing problem (Lesson 6) or toward unrelated tangents? If learners are not yet sensing the next step in the storyline, begin Lesson 6 with a more explicit bridge from scalar multiplication back to the ferry's misdirection problem. (4) Did all learners successfully add annotated elements to their cumulative ferry model, and were annotations written in complete sentences that reference the phenomenon? Underdeveloped annotations suggest learners can perform the procedure but have not connected it to meaning — consider a model annotation gallery walk at the start of Lesson 6.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners drew scaled vectors geometrically (v, 2v, −v) using ruler and scale, calculated scalar multiples of column vectors (2p, −3q, ½r), and tested four vector pairs for parallelism by finding scalar k — including identifying a non-parallel pair (e and f) where component ratios were unequal.
What did I learn?	Multiplying a vector by a positive scalar scales its magnitude while preserving its direction; multiplying by a negative scalar reverses the direction; two vectors are parallel if and only if one is a scalar multiple of the other (with the sign of k indicating same or opposite sense); these operations model real changes in the ferry scenario such as a stronger current or increased engine thrust.
How does this explain the phenomenon?	Learners explained why the Kisumu ferry would drift even further south if the lake current were scaled up (e.g., 1.5c or 2c), calculated the resulting drift distance using scalar multiplication, identified the engine's original and increased thrust vectors as parallel (same direction, different magnitude), and recorded a Captain's Advisory quantifying the increased south-drift for a doubled current.

LESSON 6 (80 minutes): Magnitude of a Vector — How Far Did the Ferry Really Travel?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To enable learners to calculate the magnitude of a vector using the Pythagorean theorem and apply this skill to determine the true distance travelled by the Kisumu ferry under the combined effect of engine thrust and lake current.
Knowledge	Learners will know that the magnitude of a vector is its numerical length, calculated using the Pythagorean theorem when the vector is expressed as a sum of perpendicular components, and that magnitude is always a non-negative scalar quantity.
Skills	Learners will be able to: (i) represent a 2-D vector as an ordered pair of perpendicular components; (ii) apply the formula $\lvert v \rvert = \sqrt{x^2 + y^2}$ to calculate the magnitude of a vector; (iii) interpret the magnitude in context as a real-world distance or speed.
Attitudes	Learners will appreciate the precision and reliability that mathematical tools provide when making navigation decisions, and will value accuracy in computation as a life-saving professional skill.
Key Inquiry Question	How can vectors be represented, combined, and applied to solve problems involving displacement, navigation, and geometry?
Purpose in Storyline	In Lessons 1–5 learners established what vectors are, how to represent them, how to add them graphically (tip-to-tail and parallelogram), and how to resolve a resultant vector into components. In Lesson 6 learners now ask: given those components, exactly how long is the resultant vector? This directly answers 'How far did the ferry actually travel?' — the magnitude question at the heart of the anchoring phenomenon — and sets up Lesson 7, where they will calculate the true direction (bearing) of that journey.
Safety Notes	No hazardous materials or equipment are used in this lesson. If grid-paper or compasses are used for drawing, remind learners to handle compass points carefully. Ensure learners do not share calculators in ways that disrupt peer work time.

B. LESSON OVERVIEW

Learners return to the Kisumu–Mbita ferry scenario armed with the vector component knowledge built in Lessons 1–5. Through a structured predict–observe–explain cycle they discover that the ferry's 20 km/h eastward engine thrust and the 8 km/h southward lake current together form a right-angled vector triangle, and that the Pythagorean theorem — already familiar from geometry — is exactly the tool needed to calculate the true distance the ferry travelled. Working in pairs and small groups, learners compute $\lvert v \rvert = \sqrt{20^2 + 8^2} = \sqrt{464} \approx 21.5$ km/h (resultant speed) and, scaling for time, find the actual displacement magnitude. They record findings on their Driving Question Boards and update their cumulative ferry-journey models, marking the resultant vector with its calculated magnitude.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Vector word problem: resultant velocity] Source: Khan	Learners are presented with a precise recap scenario on the board: 'The ferry engine pushes due east at 20 km/h. The Lake Victoria current pushes due south at 8 km/h. The two forces act	Begin by pointing to the ferry diagram from Lesson 1 still displayed on the classroom wall (or redrawn on the board): 'We have been building up our understanding of this ferry journey	Prediction Anchoring — by committing estimates to the board before instruction, learners create cognitive dissonance that motivates the upcoming calculation. This strategy	Observable indicator: every learner has a written estimate and a written sentence of reasoning in their book before the cold-call phase. The teacher scans the room during the 3-minute window;

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	simultaneously for 3 hours. Your task right now: WITHOUT calculating anything, write down your best estimate of how far the ferry actually travelled from Kisumu, and explain your reasoning in one sentence.' Learners write individually in their exercise books for 3 minutes, then share with a shoulder partner. The teacher cold-calls 4 learners to share estimates and reasoning, recording each on the board in a 'Predictions Wall' column. Common responses expected: 'It travelled 60 km east because the engine said so,' '28 km because 20 + 8 = 28,' '(20 × 3) + (8 × 3) = 84 km,' or a Pythagorean guess. The teacher does NOT correct or validate — all predictions are honoured and will be revisited at the end of the lesson.	lesson by lesson. Today we ask the question the captain most urgently needs answered — exactly how far did the ferry travel?' Distribute the prediction slip or ask learners to open to a fresh page. SAY EXACTLY: 'Do not calculate yet. I want your gut feeling and your reasoning. You have 3 minutes — go.' Set a visible timer. After 3 minutes, cold-call Learner 1: 'Tell us your estimate and why.' WAIT 12 seconds after posing the question before accepting the answer. Repeat for 3 more learners. Record all four estimates on the board under the heading 'Our Predictions — We Will Check These.' Do NOT evaluate or correct. Say: 'Hold on to your reasoning. By the end of this lesson, you will know exactly who was closest — and why the mathematics gives us certainty that intuition alone cannot.'	(Schwartz & Bransford, 1998) ensures the Pythagorean result is felt as a resolution, not an arbitrary formula.	any learner without writing is gently prompted: 'Even a guess counts — what feels right to you?' The four cold-called responses reveal the class's prior conceptions (additive vs. Pythagorean thinking) and guide emphasis during Phase 3.
Observe Phase 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Vector Addition (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive a structured observation worksheet titled 'The Ferry's Right-Angle Triangle.' The worksheet contains: (a) a pre-drawn Cartesian grid with the vector $20\hat{\imath}$ (east, horizontal) and $-8\hat{\jmath}$ (south, vertical) already plotted as two sides of a right-angled triangle; (b) the resultant vector drawn from the origin to the terminal point at (20, −8) — labelled 'r' with a question mark for its length; (c) a zoomed-in right-angle symbol at the corner where the east and south vectors meet; (d) a side-panel showing three familiar right-triangle examples with hypotenuse calculations already completed — a 3-4-5 triangle (Pythagoras), a 5-12-13 triangle, and a 6-8-10 triangle —	Distribute the worksheet. SAY: 'You have 8 minutes to work through this sheet — read it carefully, annotate the diagram, and study the three examples in the side panel. I am looking for the mathematical pattern that connects all three scaffold triangles.' Circulate. After 4 minutes, pause the class: 'Raise your hand if you have written the word hypotenuse somewhere on your sheet.' Survey the room. If fewer than half have it, redirect: 'Look at the side panel again — what is the longest side of each triangle called?' WAIT 10 seconds. Resume circulation. At 8 minutes, cold-call 2 learners: 'Read me the sentence you completed at the bottom of the sheet.' WAIT 12	Worked Example Analysis with Bridging Analogies — learners inspect completed familiar cases (3-4-5, 5-12-13, 6-8-10) before tackling the novel ferry case. This reduces cognitive load and surfaces the structural similarity between the Pythagorean theorem and vector magnitude calculation, making the new formula feel like a natural extension of prior knowledge rather than a new rule.	Observable indicator: learners correctly label the ferry vector diagram with 'hypotenuse,' 'right angle,' '20 km/h,' and '8 km/h' before phase 3. The teacher checks at least 6 books during circulation. Any learner who has not identified the right-angle corner is asked: 'In what direction is the engine pushing? In what direction is the current pushing? Can those two directions ever be the same line?' This questioning guides re-labelling without giving the answer.

	as worked scaffolds. Learners first annotate the ferry diagram: label the horizontal side '20 km/h,' the vertical side '8 km/h,' and the right-angle corner. They then examine the three scaffold examples and identify the pattern: hypotenuse = $\sqrt{(side_1^2 + side_2^2)}$. Finally, learners write in their own words: 'The resultant vector in the ferry diagram is like the _______ of a right-angled triangle.'	seconds each time. Write on the board, in learner language: 'The resultant vector IS the hypotenuse.' Underline it. Transition: 'If the resultant is the hypotenuse, and we know both shorter sides, which theorem from your geometry work tells us exactly how to find the hypotenuse?' WAIT 15 seconds. Accept cold-calls from 3 learners.		
Explain Phase 📹 **VIDEO:** [Vector word problem: resultant velocity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners now work through a three-part structured explanation task. PART A (Derive the formula): Learners write in their books — guided by teacher board work — the general formula for vector magnitude: if vector $v = x\hat{\imath} + y\hat{\jmath}$, then $\lvert v\rvert = \sqrt{(x^2 + y^2)}$. They annotate each symbol in their own words. PART B (Apply to the ferry): Learners individually calculate the resultant speed: $\lvert v\rvert = \sqrt{(20^2 + 8^2)} = \sqrt{(400 + 64)} = \sqrt{464} \approx 21.54$ km/h. They then calculate the actual displacement magnitude over 3 hours: distance = speed × time = $21.54 \times 3 \approx 64.6$ km. Learners compare this to the straight-line Kisumu–Mbita distance of 60 km (engine-only path) and to their own predictions from Phase 1. PART C (Generalise): Learners complete three further magnitude problems set in Kenyan contexts — (i) a Nairobi matatu travelling 15 km east then 9 km north; (ii) a Rift Valley farmer walking 7 km west then 24 km south to reach a water point; (iii) a Mombasa dhow moving 10 km north-east (given components $7.07\hat{\imath} + 7.07\hat{\jmath}$). For each, learners calculate the magnitude and write one sentence interpreting the answer in context.	PART A — Board work: write '$v = x\hat{\imath} + y\hat{\jmath}$' and beneath it '$\lvert v\rvert = \sqrt{(x^2 + y^2)}$'. Say: 'This formula is Pythagoras dressed in vector language. The x-component is one leg, the y-component is the other leg, and the magnitude is the hypotenuse. Copy this into your book and annotate every symbol.' Give 2 minutes. PART B — Say: 'Now apply it. Use the ferry values. Show every step — no skipping.' Give 5 minutes of silent individual work. Circulate and check: if a learner writes $\sqrt{(20 + 8)} = \sqrt{28}$, kneel to their level and ask quietly: 'Read the formula again — what operation sits between x^2 and y^2?' WAIT 10 seconds. Do NOT give the answer. After 5 minutes, cold-call 1 learner to read their full working aloud. Ask the class: 'Do you agree? Thumbs up or thumbs down — silent vote.' Address any thumbs-down responses. PART C — Say: 'Now you are the mathematician. Three Kenyan journeys, same tool. Work in pairs. You have 10 minutes.' Circulate, focusing on the dhow problem — many learners will struggle to see that the components are already given. After 10 minutes, cold-call one pair per problem (3 cold-calls). For each, ask: 'What does this	Graduated Release with Contextualised Practice — the teacher models the formula derivation (I do), learners apply it to the ferry (We check), then learners independently solve three novel Kenyan-context problems (You do). This gradual release ensures procedural fluency is built within a meaningful context rather than through decontextualised drill. Returning to the Predictions Wall at the end closes the cognitive loop opened in Phase 1.	Observable indicators: (i) In PART B, every learner's book shows the expanded form $\sqrt{(400 + 64)} = \sqrt{464}$ — not a shortcut — confirming they applied the formula correctly; (ii) In PART C, the matatu answer ≈ 17.5 km, the farmer answer = 25 km (a 7-24-25 Pythagorean triple), and the dhow answer ≈ 10 km. The teacher uses these three benchmarks as rapid diagnostic checks during circulation. Any learner with the farmer answer written as $\sqrt{(7 + 24)} = \sqrt{31}$ has a formula application error and needs targeted re-teaching of 'squaring before adding.'

		magnitude tell us in real life?' WAIT 12 seconds. Close Part C by returning to the Predictions Wall: 'Look at our four predictions from the start. Who was closest? Why did adding 20 and 8 not give the right answer?' Cold-call 2 learners. SAY: 'The square root of the sum of squares — not the sum itself — is what the mathematics demands. That is why direction matters so deeply in vector work.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Vector word problem: resultant velocity] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Each learner pair receives two sticky notes. On the first (green), they write one thing they can now answer about the ferry's journey that they could not answer at the start of the lesson. On the second (orange), they write one remaining question about the ferry's journey — something they still want to know. Pairs walk to the class Driving Question Board and post their notes in the correct columns: 'We Now Know' (green) and 'We Still Wonder' (orange). The class does a 2-minute silent gallery walk of all new postings. The teacher then reads 3 selected 'We Still Wonder' notes aloud and says: 'We know the ferry travelled approximately 64.6 km. We know how far. But do we know in which exact direction it landed? That is our next question — and it is the question Lesson 7 will answer.' The teacher physically circles the 'direction/bearing' sticky notes on the DQB.	Distribute sticky notes. SAY: 'You now know how far the ferry really travelled. Update the board — be specific. Do not write 'I learned Pythagoras.' Write what the magnitude of the ferry's resultant vector actually is and what it means for the passengers.' Give 3 minutes for writing. Supervise the posting — ensure learners read at least two other pairs' green notes before returning to their seats. Read 3 orange 'wonder' notes aloud: choose one that mentions direction, one that mentions how to fix the ferry route, and one that raises any new idea. SAY: 'These orange notes are not gaps — they are the next steps in our mathematical journey. We are building toward the full answer to our driving question. Today we answered: how far. Lesson 7 answers: in which direction.'	Driving Question Board — Cumulative Knowledge Tracking: the DQB functions as a public, living document of the class's growing understanding. By requiring specific rather than generic green-note responses, the teacher ensures learners articulate the lesson's core concept in their own words, reinforcing retention. The orange notes preview Lesson 7 authentically, so the next lesson begins with genuine student-generated curiosity rather than a teacher-imposed transition.	Observable indicator: green sticky notes must reference either a number (e.g., '≈ 64.6 km' or '21.5 km/h') or the words 'magnitude' and 'Pythagorean theorem' to be considered complete. Any note that only says 'vectors add' or 'Pythagoras is used' is flagged by the teacher during the gallery walk with a quiet prompt: 'Can you be more specific — what number did you calculate for the ferry?'
Model Building Phase 📹 **VIDEO:** [Vector word problem: resultant velocity]	Learners return to their cumulative ferry-journey model — a scaled vector diagram of the Kisumu–Mbita scenario that has been updated in each previous lesson. Today's task: annotate the resultant vector arrow with its	Say: 'Take out your ferry model diagram from your portfolio. Today you are adding the magnitude — the numerical length — to your resultant vector. Use your ruler to measure the arrow you drew last lesson and check it against your	Cumulative Model Revision — each lesson adds one layer of mathematical detail to the same ferry diagram, making the model increasingly precise and complete. This strategy makes the unit's progression visible and tangible:	Observable indicator: the model update statement must include (i) the numerical magnitude ≈ 21.5 km/h, (ii) the calculated distance ≈ 64.6 km, and (iii) an acknowledgement that direction is still missing. The teacher collects 6

Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Vector Addition (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	calculated magnitude. Using a ruler and the scale established in earlier lessons (e.g., 1 cm = 5 km/h), learners measure the resultant arrow they drew in Lesson 5 and verify that its physical length matches the calculated magnitude of ≈ 21.5 km/h (≈ 4.3 cm at the given scale). They write alongside the arrow: '$\|v\| = \sqrt{(20^2 + 8^2)} = \sqrt{464} \approx 21.5$ km/h' and add a small legend box to their diagram: 'Magnitude = length of resultant vector.' Learners also annotate the two component arrows with their magnitudes (20 km/h east, 8 km/h south). Finally, learners write one 'model update statement' at the bottom of their diagram page: 'My model now shows that the ferry's resultant velocity has a magnitude of approximately 21.5 km/h, meaning it travelled about 64.6 km in 3 hours — not 60 km as the engine alone would suggest. I still need to add the exact direction to complete the captain's navigation tool.'	calculation.' Give 5 minutes. Circulate and check that: (i) the label on the resultant arrow matches the calculated value; (ii) the scale verification is attempted; (iii) the model update statement is written. After 5 minutes, cold-call 2 learners to read their model update statements aloud. Ask the class: 'What is still missing from our model that would complete the captain's navigation tool?' WAIT 15 seconds. Accept cold-calls from 3 learners. Expected answer: the direction/bearing. SAY: 'Your model is 80% complete. The magnitude is there. The direction is missing. When that is added in Lesson 7, the captain will have a complete mathematical navigation instruction — and we will be very close to fully answering our driving question.' Point to the driving question displayed on the wall and read it aloud together as a class.	learners can literally see their growing understanding represented in the annotations on a single diagram. The scale-verification step also integrates the graphical and algebraic representations, deepening conceptual understanding of vector magnitude.	models at the end of the lesson (rotating collection so all models are reviewed over the unit) and provides written feedback before the next lesson. Any model missing the scale-verification step is noted for follow-up at the start of Lesson 7.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners in Phase 1 predominantly use additive thinking (20 + 8 = 28) or multiplicative/Pythagorean thinking? What does this tell you about the prior-knowledge gaps you need to address before Lesson 7? (2) In Phase 3 Part C, which of the three Kenyan-context problems caused the most difficulty — and why? (3) Did the Predictions Wall in Phase 1 genuinely create cognitive dissonance, or did learners forget their predictions by Phase 3? If the latter, consider reading predictions aloud again at the start of Phase 3 in future iterations. (4) Were learners able to interpret the magnitude (64.6 km) in the context of the ferry scenario, or did they treat it as an abstract number? If interpretation was weak, plan a brief context-reconnection routine at the start of Lesson 7. (5) Review the 6 collected models: are annotations mathematically precise and contextually meaningful, or generic? Use findings to differentiate the opening of Lesson 7.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	The ferry departed Kisumu heading due east at 20 km/h while a southward Lake Victoria current of 8 km/h simultaneously pushed it off course. The resultant journey formed a right-angled triangle with legs of 20 km/h (east) and 8 km/h (south).
What did I learn?	The magnitude of a resultant vector formed from two perpendicular components is calculated using the Pythagorean theorem: $\|v\| =$

	$\sqrt{(x^2 + y^2)}$. For the ferry, $\|v\| = \sqrt{(20^2 + 8^2)} = \sqrt{464} \approx 21.5$ km/h, meaning over 3 hours the ferry actually travelled approximately 64.6 km — not the 60 km straight-line engine distance.
How does this explain the phenomenon?	Because the engine thrust (east) and the current (south) are perpendicular directed quantities, they combine as the two legs of a right-angled triangle. The actual path of the ferry is the hypotenuse of that triangle. Pythagoras gives us the exact length of the hypotenuse — and therefore the true distance travelled. This is why simply adding 20 and 8 is mathematically wrong: direction forces us to use squares and square roots, not plain addition.

LESSON 7 (80 minutes): Collinear Points and Vector Paths	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To determine whether points are collinear using vectors and to describe paths between multiple points using vector chains, enabling students to plan and verify multi-stop routes on Lake Victoria.
Knowledge	Students know that three or more points are collinear if the vectors connecting them are parallel (scalar multiples of each other) and share a common point; they know that a vector path between multiple points can be expressed as a chain of position vectors or displacement vectors.
Skills	Students can test for collinearity of three points using vectors; express multi-point journeys as vector chains; simplify vector paths using vector addition; and interpret collinearity results in navigational and geometric contexts.
Attitudes	Students appreciate the precision and reliability that mathematical reasoning brings to real-world navigation, showing commitment to accuracy when lives and livelihoods depend on correct positioning.
Key Inquiry Question	How can vectors be represented, combined, and applied to solve problems involving displacement, navigation, and geometry?
Purpose in Storyline	In previous lessons, students learned to represent vectors, add them using the triangle and parallelogram laws, calculate resultant magnitudes using Pythagoras, and find directions using trigonometry. Now, in Lesson 7, students confront a new navigational question: if the ferry makes multiple stops — say Kisumu → Kendu Bay → Mbita — do those three ports lie on a straight-line route? Can the captain plan a straight-path multi-stop journey, and how does vector mathematics confirm or deny that such a straight path exists? Collinearity and vector chains are the tools that answer this, directly advancing the driving question toward its final resolution in Lesson 8.
Safety Notes	No physical hazards in this lesson. Remind students to use rulers and set squares carefully when drawing vector diagrams on grid paper to avoid inaccurate constructions that could lead to incorrect conclusions.

B. LESSON OVERVIEW
In this 80-minute lesson, students investigate collinear points and multi-stop vector paths in the context of the Lake Victoria ferry scenario. Building on their established knowledge of vector representation, addition, magnitude, and direction from Lessons 1–6, students now ask: can three ferry ports lie on a perfectly straight route, and how does a captain mathematically verify this before setting sail? Students first predict whether three given ports are collinear by visual inspection, then gather evidence by computing vectors between pairs of points and testing for scalar multiples. They apply this understanding to express multi-stop ferry journeys as vector chains and simplify them. By the end of the lesson, students can confidently test collinearity using vectors and construct vector path chains — tools they will use in Lesson 8 to fully equip the Kisumu ferry captain with a complete navigational mathematical toolkit.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Terms & labels in	Students are presented with a printed or on-screen coordinate grid showing three labelled ports:	Display the coordinate grid on the board. Say exactly: 'Look at the three ports on this map of Lake	Think-Pair-Share with recorded predictions. This activates prior intuition about straight lines and	Observable indicator: Students can articulate a geometric reason (e.g., 'they look like they are on the

[geometry](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	K (Kisumu) at position (0, 0), B (Kendu Bay) at (4, 2), and M (Mbita) at (8, 4). They are asked: 'Without doing any calculation, do you think these three ports lie on a perfectly straight route — one the ferry could travel in a single straight line with intermediate stops? Write your prediction and your reason in your exercise book.' Students write individual predictions (approximately 3 minutes), then share with a neighbour (Think-Pair-Share, 2 minutes).	Victoria. Kisumu is at the origin. Kendu Bay is here. Mbita is here. If you were the captain, could you draw one perfectly straight line through all three? Write yes or no — and most importantly, write WHY you think so. You have 3 minutes of silent thinking.' [Allow 3 minutes silent writing. Do NOT confirm or deny any prediction.] After Think-Pair-Share, cold-call 3 students: one who said YES, one who said NO, and one who was unsure. Ask each: 'What was your reasoning?' Record key phrases from each on the board under the heading 'Our Predictions.' Say: 'By the end of today, you will have a mathematical tool that proves — not just shows — whether three points are collinear. Let's build that tool.'	collinearity, creating a cognitive hook. Recording predictions publicly on the board creates accountability — students are motivated to see whether their geometric intuition matches the algebraic result, mirroring the kind of verification a real navigator must perform.	same line' or 'the spacing looks equal') even before formal instruction. Listen for students who reference gradient or equal spacing — these show readiness for the algebraic method. Flag students who have no reasoning at all for targeted support during Phase 2.
Observe Phase 📹 **VIDEO:** [Terms & labels in geometry](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Matrix Multiplication: Vectors (PLIX)](#) Source: CK-12 🔍 Search ARES for similar readings	Students work in pairs to gather vector evidence. Using the three ports K(0,0), B(4,2), M(8,4), each pair: (1) Calculates vector KB = B − K and vector KM = M − K. (2) Checks whether KM = n × KB for some scalar n. (3) Draws both vectors on their grid, starting from K, and observes whether they lie along the same straight line. They then receive a second set of points — ports P(1,3), Q(3,7), R(5,10) — and repeat the test. Students record their vector calculations and conclusions in a structured evidence table with columns: [Points | Vector 1 | Vector 2 | Scalar multiple? | Collinear?]. A short worked example of scalar multiple testing (using a Nairobi–Nakuru–Eldoret road analogy) is displayed on the board for reference.	Distribute grid paper and the structured evidence table (pre-printed or drawn on board for copying). Say: 'Your job right now is to be mathematical detectives. You are going to calculate two vectors from Kisumu — one to Kendu Bay, one to Mbita — and then ask: is one of these vectors just a stretched or shrunk version of the other? That relationship has a name: scalar multiple. If vector KM equals some number times vector KB, then those points are collinear.' [Allow 12 minutes for pair work on the first set of points. Circulate — observe written calculations, not just answers.] After 12 minutes, cold-call 2 pairs to share their working for the first set. Say: 'Show us your calculation step by step — not just the answer.' [Allow 10 seconds WAIT TIME before accepting responses.] Say: 'Now try the second set of	Structured Evidence Table with Guided Inquiry. Naming the sensemaking strategy: Comparative Case Analysis. Students investigate two cases — one collinear, one non-collinear — side by side. Comparing the two cases makes the defining feature (scalar multiple relationship) visible by contrast, which is far more powerful than studying a single example in isolation. The Nairobi–Nakuru–Eldoret road analogy grounds the abstract idea in a familiar Kenyan context of straight-road travel.	Observable indicators: (1) Students correctly compute KB = (4,2) and KM = (8,4) and identify KM = 2·KB, concluding collinearity. (2) For P, Q, R: students compute PQ = (2,4) and PR = (4,7); PR ≠ 2·PQ (since (4,7) ≠ 2×(2,4) = (4,8)), so P, Q, R are NOT collinear — students who reach this conclusion correctly are on track. (3) Watch for the common error of testing only magnitudes without checking direction components separately — address this explicitly if observed in more than 2 pairs.

		ports — P, Q, R. Are these three ports collinear? Use the same method. You have 8 minutes.' [Circulate and listen for students who test only one pair of vectors — prompt: 'Is it enough to check only KB and KM? What if you also checked BM?'] Cold-call 1 pair for the second set. Ask: 'What was your conclusion, and how confident are you in it?'		
Explain Phase 📹 **VIDEO:** [Terms & labels in geometry] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students consolidate the collinearity test into a formal definition in their own words, then immediately extend it to vector chains. The teacher introduces the concept of a vector path: if the ferry travels Kisumu → Kendu Bay → Homa Bay → Mbita, the total displacement can be expressed as a chain: KM = KB + BH + HM. Students work individually on two tasks: (Task A) Given points A(2,1), B(5,7), C(8,13) — prove algebraically that A, B, C are collinear. (Task B) A ferry travels from Kisumu K(0,0) to Kendu Bay B(3,2), then to Homa Bay H(7,5), then to Mbita M(10,8). Express the total journey as a vector chain and calculate the single resultant displacement vector KM. Students then check: are K, B, H, M collinear?	Open the Explain phase by asking the class: 'In your own words — not mine — what is the test for collinearity using vectors? Write one sentence.' [Allow 2 minutes silent writing. WAIT TIME: 15 seconds before cold-calling.] Cold-call 3 students. Write the best student-generated definition on the board with the student's name: 'Our class definition.' Then say: 'Now I want to introduce the idea of a vector chain — or vector path. Think about the ferry making multiple stops. Each leg of the journey is a separate vector. The total journey is all those vectors added end to end. This is called a vector chain.' [Write the general form on the board: Total vector = v1 + v2 + v3 + ...] Say: 'Task A is about proving collinearity. Task B is about building and simplifying a vector chain — and then using your collinearity test on the result. Work individually. You have 15 minutes.' [Circulate — for Task A, look for students who correctly compute AB = (3,6) and AC = (6,12) and identify AC = 2·AB. For Task B, look for students who correctly add (3,2)+(4,3)+(3,3) = (10,8) and then test KB, KH, KM for collinearity.] After 15 minutes, facilitate a whole-class discussion: 'What did you find for Task B — is	Student-Generated Definition followed by Worked Application (Gradual Release). Naming the sensemaking strategy: Formalise–Apply–Reflect. Students first formalise the idea in their own language (personalising abstract knowledge), then immediately apply it in a novel navigational context (Task B), then reflect on whether the result is meaningful for the ferry captain. This sequence ensures conceptual understanding precedes procedural fluency — consistent with CBE's emphasis on competence over rote recall.	Observable indicators: (1) Task A: Students show AB = (3,6), AC = (6,12), state AC = 2·AB, and write a clear conclusion: 'A, B, C are collinear.' (2) Task B: Students correctly compute the vector chain sum as (10,8) and test for collinearity of K, B, H, M. Award particular credit to students who recognise that if all intermediate points produce scalar multiples of the same base vector, the entire route is a straight line. (3) Misconception to watch: students who think a vector chain can only have two terms — prompt with: 'How many stops does the ferry make? How many vector arrows do you need?'

		this a straight-line route for the ferry captain?' [WAIT TIME: 10 seconds.] Cold-call 2 students and ask them to justify their answer at the board.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Terms & labels in geometry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students return to the class Driving Question Board. They revisit the original question: 'Why did the Kisumu ferry miss Mbita — and how can mathematics give the captain the tools to always reach the right destination, no matter the current or wind?' Each student writes one sticky note (or a written card) responding to the prompt: 'What new mathematical tool did we gain today, and how does it help the captain plan a safe, correct route?' Students also update their personal DQB trackers — a two-column table titled 'What we now know' and 'What we still need to figure out.' The class notes that Lesson 8 (the final lesson) will synthesise all tools into a complete navigational plan for the captain.	Direct students to the DQB. Say: 'Look at the question that started our entire unit. We have now spent seven lessons building a mathematical toolkit for that captain. Today we added two new tools: the collinearity test and the vector chain. Write on your sticky note: which of these tools helps the captain most, and why?' [Allow 4 minutes for individual writing. WAIT TIME before collecting.] Ask 3 students to read their sticky notes aloud and place them on the DQB. Then ask the whole class: 'What do we still NOT know that the captain needs?' [Accept 2–3 responses. Write unresolved questions in a designated 'Still wondering' column on the DQB.] Say: 'In Lesson 8, we bring everything together. By the end of that lesson, you will be able to give the captain a complete mathematical answer to every part of our driving question.'	Driving Question Board Revisit with Personal Tracker Update. Naming the sensemaking strategy: Cumulative Storyline Mapping. Students explicitly connect today's new tools to the growing body of evidence accumulated across 7 lessons. By naming both what they now know and what remains unknown, students develop metacognitive awareness of their own learning trajectory — a core CBE competency. The public DQB also creates shared intellectual accountability across the class.	Observable indicators: Students' sticky notes should explicitly name 'collinearity test' or 'vector chain' and connect it to the ferry scenario (e.g., 'The captain can check if three ports are on a straight path before sailing'). Students who write only a general statement ('we learned about vectors') without connecting to the phenomenon need additional prompting. The 'Still wondering' column should surface at least one genuine unresolved question (e.g., 'What if the route is not straight at all?' or 'How does the captain adjust for current on a multi-stop journey?') — these questions feed directly into Lesson 8.
Model Building Phase 📹 **VIDEO:** Terms & labels in geometry Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX)	Students revise and extend their cumulative vector model of the Lake Victoria ferry scenario. They began this model in Lesson 1 (drawing a simple arrow for velocity) and have progressively added: vector addition diagrams, resultant vectors, magnitude calculations, direction angles, and position vectors. Today they add to their model: (1) A collinearity test diagram — showing Kisumu, Kendu Bay, and Mbita with labelled vectors and a written proof of collinearity. (2) A multi-stop	Say: 'Take out your cumulative vector model — the one you have been building since Lesson 1. Today you are adding two new layers. First, draw and label a collinearity proof for Kisumu, Kendu Bay, and Mbita. Second, draw the full four-port vector chain and the resultant displacement arrow. Add a written caption explaining what these two additions mean for the captain.' [Allow 12 minutes for model revision. Circulate and observe: Are vector arrows drawn with	Cumulative Model Revision with Gallery Walk Peer Review. Naming the sensemaking strategy: Progressive Model Elaboration. Each lesson adds a new representational layer to the same model, making the growth of understanding visible and tangible. The Gallery Walk introduces peer accountability and exposes students to alternative representational choices — building the critical-evaluation competency that CBE emphasises. Peer feedback ('one	Observable indicators: (1) Students' collinearity diagram correctly shows KB = (4,2), KM = (8,4), with the scalar multiple KM = 2·KB written alongside and a clear written conclusion. (2) The vector chain diagram shows four sequential arrows from K to B to H to M, correctly labelled, with the resultant KM = (10,8) drawn as a single arrow from K to M. (3) During Gallery Walk, peer sticky notes should identify specific features of the model (e.g., 'Your scalar multiple calculation is clear'

Source: CK-12 🔍 Search ARES for similar readings	vector chain diagram — showing all four ports (Kisumu, Kendu Bay, Homa Bay, Mbita) connected by sequential vector arrows, labelled with component vectors, and a resultant displacement arrow drawn from Kisumu directly to Mbita. Students annotate their model with the caption: 'The captain can verify a straight-line multi-stop route using the scalar multiple test, and can calculate total displacement using a vector chain — without needing to re-sail each leg separately.'	correct direction and labelled with component notation? Are scalar multiple calculations written alongside the diagram?] After 12 minutes, ask pairs to do a 'Gallery Walk' — each pair reviews the model of one other pair and writes one specific strength and one specific question on a sticky note left with that model. [Allow 4 minutes.] Close by asking: 'How has your model of the ferry journey grown from Lesson 1 to today? What does it show now that it could not show then?' Cold-call 2 students. [WAIT TIME: 12 seconds.]	strength, one question') is structured to be specific and constructive rather than evaluative.	or 'I notice your resultant arrow does not start from K — should it?'). Students who have not connected their model additions to the ferry scenario caption need teacher follow-up before Lesson 8.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students successfully distinguish between testing collinearity visually (prediction) versus algebraically (evidence)? Note the names of any students who remained uncertain about the scalar multiple test — these students will need a quick individual check-in at the start of Lesson 8. (2) In the Explain phase, did students correctly identify that K, B, H, M in Task B are collinear — and did they explain what this means for the ferry captain? If fewer than 70% of students reached a correct conclusion for Task B, consider opening Lesson 8 with a 5-minute revisit of vector chain simplification before moving to the synthesis activity. (3) Were the DQB sticky notes specific enough? Did students name the new tools (collinearity test, vector chain) and connect them to the phenomenon, or did they write generically? If DQB entries were vague, adjust the Lesson 8 DQB prompt to require students to use at least two specific mathematical terms. (4) During the Gallery Walk, was the peer feedback constructive and specific? If pairs struggled to write a genuine 'question' for their peers' models, model exemplar feedback language at the start of Lesson 8. (5) Overall storyline check: students have now gathered evidence across 7 lessons. They can represent, add, scale, find magnitudes and directions of vectors, use position vectors, and test collinearity. Does the class feel ready to synthesise all of these tools in Lesson 8 into a complete response to the driving question? If not, identify the specific gap and address it in the Lesson 8 opening.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed that when three ports — Kisumu K(0,0), Kendu Bay B(4,2), and Mbita M(8,4) — are plotted on a coordinate grid, the vectors KB and KM appear to lie along the same straight line. They also observed that for a second set of ports P(1,3), Q(3,7), R(5,10), the visual alignment is less obvious, requiring algebraic confirmation.
What did I learn?	Students learned that three points are collinear if and only if the vectors connecting them from a common point are scalar multiples of each other — meaning one vector is a stretched or shrunk version of the other in exactly the same direction. They also learned that a multi-stop ferry journey can be expressed as a vector chain (the sum of sequential displacement vectors), and that this chain simplifies to a single resultant displacement vector from start to finish.
How does this explain the phenomenon?	Students explained that the Kisumu ferry captain can use the scalar multiple collinearity test to verify before sailing whether a planned multi-stop route is truly a straight-line path, avoiding unnecessary course corrections. They also explained that by expressing the full journey as a vector chain, the captain can calculate the exact total displacement from port of departure to final

	destination — without needing to measure each leg separately — giving precise, mathematically guaranteed navigational information.

LESSON 8 (80 minutes): Translation Vectors and Unit Synthesis — Solving the Ferry Problem	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate all vector concepts learned in the unit by applying translation vectors and resultant vector analysis to fully and precisely solve the anchoring Kisumu ferry phenomenon, demonstrating mastery of the key inquiry questions.
Knowledge	Students know that: (1) a translation vector describes a movement from one point to another in the plane with both magnitude and direction; (2) vectors can be represented in column vector form and on Cartesian grids; (3) the resultant of two or more vectors is found by vector addition (triangle or parallelogram law); (4) the magnitude of a resultant vector is found using Pythagoras' theorem; (5) the direction of a resultant vector is found using trigonometric ratios (tan θ); (6) position vectors locate points relative to an origin; (7) scalar multiplication scales a vector without changing its direction (unless the scalar is negative).
Skills	Students are able to: (1) represent real-world displacement situations as column vectors; (2) add vectors graphically (tip-to-tail) and algebraically; (3) calculate the magnitude and true bearing/direction angle of a resultant vector; (4) apply translation vectors to map geometric transformations on the Cartesian plane; (5) construct and compare vector models at different levels of sophistication; (6) communicate mathematical reasoning using precise vector language.
Attitudes	Students demonstrate: perseverance in revisiting and refining their models across the unit; confidence in using mathematics to explain real-world phenomena; appreciation that precision in direction and magnitude saves lives and livelihoods (as with the Kisumu ferry passengers and traders); collaborative responsibility in synthesising shared evidence from the Driving Question Board.
Key Inquiry Question	How can vectors be represented, combined, and applied to solve problems involving displacement, navigation, and geometry? Why is direction essential, and how does the full vector toolkit — addition, magnitude, direction, translation — give the captain the tools to always reach the right destination?
Purpose in Storyline	This is the culminating lesson of the unit. Students have spent seven lessons building the conceptual and procedural toolkit: scalar vs. vector distinction, vector representation, addition laws, magnitude, direction, position vectors, and scalar multiplication. In Lesson 8, all of these threads are woven together. Students explicitly solve the Kisumu–Mbita ferry problem with full mathematical precision — calculating the resultant displacement, the true distance travelled, and the corrective heading the captain needed. They compare their Lesson 1 naive model (a simple eastward arrow) with their final sophisticated model, complete the Driving Question Board, and articulate a Final Explanation. The ferry is no longer mysterious — it is fully explained by the mathematics.
Safety Notes	No physical hazards in this lesson. Ensure graph paper, rulers, and protractors are handled carefully. Emotional safety: affirm that Lesson 1 predictions were expected to be incomplete — growth of understanding is the goal of the unit, not being right on Day 1.

B. LESSON OVERVIEW
In this 80-minute final lesson, students synthesise the full Vectors I unit by definitively solving the anchoring phenomenon: Why did the Kisumu ferry miss Mbita, and how can the captain correct course? The lesson opens with a structured DQB Revisit in which student pairs audit every driving question posted across the unit and mark which questions can now be answered with evidence. Students then work through a Final Ferry Problem — a rich multi-part task in which they represent the engine velocity and current as column vectors, add them to find the resultant displacement vector, calculate the true distance travelled using Pythagoras, determine the drift angle using trigonometry, and design the corrective heading the captain should have set. A Lesson 1 vs. Lesson 8 Model Comparison makes the learning journey visible and concrete. Students then articulate a written Final Explanation answering both key inquiry questions. The lesson closes with DQB Completion, a whole-class celebration of questions answered, and a reflection prompt that links vector mathematics to real Kenyan navigation contexts (Lake Victoria ferries, Nairobi air traffic, Rift

Valley road surveying).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students are given back their Lesson 1 individual prediction drawings — a sketch they made in the very first lesson showing what they thought happened to the ferry. Working individually for 4 minutes, they annotate their original drawing with the question: 'What did I not know in Lesson 1 that I now know?' They circle gaps, add labels, and write 1–2 sentences describing what was missing from their early thinking. They then share with a partner using a Think-Pair-Share. The class briefly hears 3–4 voices describing the single biggest conceptual gap they have closed across the unit. This activates prior knowledge, creates cognitive contrast between naive and expert thinking, and sets a purposeful frame: today we finish the story.	Return Lesson 1 prediction sheets (prepared in advance). Say: 'Before we do anything else today, I want you to look at what you drew in Lesson 1 — your very first idea about the ferry. You had no vector tools at all. Annotate it: what did you not yet know that you now know? You have 4 minutes — go.' Set a visible 4-minute timer. Circulate silently; do NOT prompt — allow productive struggle and self-assessment. After 4 minutes: 'Turn to your partner. Take 90 seconds each. What is the single biggest idea you were missing in Lesson 1?' Allow 3 minutes of pair talk. Cold-call EXACTLY 4 students: 'Achieng — what was missing for you?', 'Otieno — something different?', 'Wanjiku — add to that?', 'Baraka — anything we haven't named yet?' Record their responses as a list on the board under the heading: 'What we have learned.' Do NOT correct or elaborate yet — this is activation and honouring student voice. Close with: 'Every single one of those gaps — we are going to close today, completely, using the full power of vectors.'	Strategy: Annotation of Prior Work. Students physically mark up their own Lesson 1 artefacts, making learning growth visible and tangible. This strategy builds metacognitive awareness — learners see themselves as knowers whose understanding has evolved — which is motivating and consolidating. It also surfaces any remaining misconceptions before the final task.	Observable indicator: Students can name at least two specific mathematical concepts (e.g., 'I didn't know vectors have direction AND magnitude', 'I didn't know you add vectors tip-to-tail', 'I didn't know how to use Pythagoras for the resultant') that were absent from their Lesson 1 model. Teacher listens for vague responses ('I knew less') and probes: 'Can you be more specific — which mathematical idea was missing?' Flag any student who cannot name a concept for targeted support during Phase 3.
Observe Phase 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English	The full Driving Question Board (DQB) — populated across all eight lessons — is displayed prominently. Student pairs receive a printed DQB Audit Sheet listing every question card on the board. Working in pairs (5 minutes), they	Point to the DQB and say: 'This board has been our scientific and mathematical notebook for the whole unit. Every question you ever asked about the ferry lives here. Today, you are the auditors.' Distribute Audit Sheets. Say: 'With	Strategy: Driving Question Board Audit (Evidence Mapping). Students map their accumulated evidence against open questions, practising the scientific practice of 'Using Mathematics and Computational Thinking' and	Observable indicator: Pairs can correctly classify at least 80% of DQB cards as green (answerable). Teacher notes any card about vector addition, resultant magnitude, or direction that is still amber/red — these indicate gaps

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	use a simple traffic-light code: GREEN = we can now answer this fully with evidence from our lessons; AMBER = we can partially answer this; RED = this question is still open. Pairs then report out using a Gallery Walk format: each pair sticks a coloured dot next to each DQB question card. The class can then see the collective picture of their understanding. Teacher facilitates a 3-minute whole-class scan: 'Look at our board. Most green — that is the evidence of your work over eight lessons. Any amber or red cards become our final targets for today.' This gives students agency over the lesson's closing agenda and affirms collective intellectual progress.	your partner, traffic-light every question. GREEN = fully answered with evidence. AMBER = partially answered. RED = still open. Five minutes — go.' Circulate and observe dot placement. After Gallery Walk, stand at the board and narrate: 'I see mostly green — fantastic. I notice [name 1–2 amber/red cards]. Let's make sure we close those today.' If the two key inquiry questions ('How do vectors differ from scalars?' and 'How can vectors be combined to solve navigation problems?') are not yet fully green, point to them explicitly: 'These two are our unit's biggest questions. By the end of today, both are green — guaranteed.' Allow 10 seconds of wait time after pointing to red/amber cards before moving on, inviting any student to speak if they wish.	'Obtaining, Evaluating, and Communicating Information.' This makes the structure of the unit visible and gives students ownership of what remains to be resolved, creating authentic motivation for the final task.	that the Phase 3 Ferry Problem must explicitly address. Teacher mentally flags those pairs for targeted cold-calling in Phase 3.
Explain Phase 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students receive the Final Ferry Problem worksheet — the most complete and mathematically rigorous version of the anchoring phenomenon they have encountered. The problem is structured in five parts: (A) Represent the engine velocity vector and the current vector as column vectors. (B) Add the two vectors to find the resultant displacement vector after 3 hours. (C) Calculate the magnitude of the resultant displacement using Pythagoras' theorem and state how far south of Mbita the ferry lands. (D) Calculate the drift angle below the eastward heading using trigonometry (tan θ) and express the true direction as a bearing. (E) Design the corrective heading: if the captain wants to arrive exactly at Mbita, what direction (as a	Distribute the Final Ferry Problem worksheet. Say: 'This is it — the problem we have been building toward for eight lessons. Every tool you need is in your notes. Parts A through D — work in your group of three. Part E — each person writes their own solution. You have 18 minutes. I want to see column vectors, a drawn diagram, Pythagoras working, and a trig calculation. Go.' Set a visible 18-minute timer. Circulate using the following targeted moves: (1) If a group is stuck on Part A, prompt: 'What does it mean to write a vector in column form? Which direction is positive x? Positive y?' WAIT 12 seconds. (2) If a group completes Part D quickly, extend: 'What bearing does that angle correspond to? Remember — bearings are measured clockwise	Strategy: Scaffolded Multi-Part Problem Solving (Progressive Complexity). The five-part structure mirrors the NGSS practice of 'Using Mathematics and Computational Thinking' — students move from representation (A), to computation (B, C, D), to design/application (E). Part E in particular requires reverse engineering: given the desired outcome, determine the required input. This is the highest level of application in the unit and directly answers the driving question ('how can mathematics give the captain the tools to always reach the right destination?'). The group-then-individual structure ensures collaborative sense-making followed by individual accountability.	Observable indicators: (1) Column vectors are correctly written with the current vector showing a negative y-component, e.g., [0, −8], confirming understanding of direction sign conventions. (2) Pythagoras is applied correctly to the components of the resultant vector. (3) The drift angle calculation uses tan θ = opposite/adjacent = 8/20, not inverted. (4) Part E: students recognise that the corrective heading requires the engine's y-component to equal +8 km/h (northward) to cancel the current — demonstrating genuine understanding of vector decomposition and addition, not just formula application. Flag any student who uses only magnitude without direction in Part E as needing conceptual reinforcement.

	bearing) must the engine be pointed to counteract the current? Students work in groups of 3 for 18 minutes, then each student independently writes a solution to Part E for individual accountability. Groups share solutions via a whole-class Structured Discussion. The teacher annotates a class solution on the board in real time. Specific numbers: engine vector = (60, 0) km over 3 hours (or velocity 20 km/h east = column vector [20, 0]); current vector = [0, −8] km/h; resultant velocity = [20, −8]; magnitude = $\sqrt{(20^2+8^2)}$ = $\sqrt{(400+64)}$ = $\sqrt{464} \approx 21.5$ km/h; drift angle = arctan(8/20) ≈ 21.8° south of east; true bearing ≈ 111.8°. For the corrective heading (Part E): the captain must angle the engine north of east so that the northward component of engine velocity exactly cancels the southward current of 8 km/h. If engine speed is 20 km/h, the northward component needed = 8 km/h, so $\sin\theta = 8/20 = 0.4$, $\theta \approx 23.6°$ north of east, bearing ≈ 066.4°.	from north.' (3) For Part E, do NOT give hints — let students struggle productively for at least 5 minutes. After 18 minutes, say: 'Pens down on group work. Part E — write your own solution now. Three minutes, independently.' After Part E, facilitate whole-class discussion: cold-call one group per part. For Part E specifically, cold-call 3 students with DIFFERENT approaches and ask the class to vote on which is correct. Annotate the agreed solution on the board. Say: 'The captain needed to aim at bearing 066° — not due east. That is the power of vectors. That is why this mathematics matters.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX)	With the full solution on the board, the class returns to the DQB for a final 8-minute Closure Session. The teacher reads each remaining amber or red question aloud. Student volunteers — not the teacher — provide the answer, pointing to specific work from the lesson or the unit. The class votes (thumbs up/thumbs down) on whether each answer is sufficient to turn the card green. Once all cards are green, the teacher asks each student to independently write a response to the two key inquiry questions in their exercise books — a paragraph for each —	Return to the DQB. Say: 'Let's go through every card that isn't fully green. I am not going to answer these — you are. Raise your hand if you can answer this one.' For each amber/red card, cold-call a student, then ask a second student: 'Do you agree? Can you add anything?' Use the class vote (thumbs up = sufficient answer, thumbs sideways = needs more, thumbs down = not answered). Do NOT accept vague answers — probe with: 'Can you say that using the word vector?' or 'Can you connect that back to the ferry?' Once all cards are green, say:	Strategy: Final Explanation Writing with Sentence Frames. Writing a Final Explanation is a culminating NGSS practice — 'Constructing Explanations.' The sentence frames scaffold without replacing student thinking, ensuring all learners can access the writing task regardless of English proficiency level. The peer-underlining move adds a low-stakes accountability layer and surfaces strong scientific language for the class to celebrate. This strategy directly answers both key inquiry questions in student voice.	Observable indicators: (1) Student Final Explanations explicitly use the words 'magnitude,' 'direction,' 'resultant,' and 'vector addition.' (2) The explanation connects abstractly named concepts back to the ferry context (not just generic definitions). (3) Students identify a specific corrective action the captain can take — not just describe the problem. Teacher collects exercise books at the end of class for written formative assessment review. Any explanation that only describes the problem without prescribing a solution (i.e., stuck at 'the ferry

Source: CK-12 🔍 Search ARES for similar readings	in their own words. This is the Final Explanation artefact. Students are given a sentence frame to support but not constrain: 'Vectors differ from scalars because... This matters for the ferry because... Vectors can be combined by... The captain can always reach the right destination by...' Students write for 6 minutes in silence.	'Every question on this board — answered. By you. That is eight lessons of mathematical work. Now — open your exercise books. I want your Final Explanation. Use the sentence frame on the board, but make it yours. Six minutes of silence.' Set timer. Circulate and read over shoulders. After 6 minutes: 'Exchange books with your partner. Read their explanation. Underline one sentence you think is particularly strong.' Allow 2 minutes of peer reading.		drifted') indicates the student has not yet reached the application level of the key inquiry question.
Model Building Phase 📹 **VIDEO:** Vector word problem: resultant velocity Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Matrix Multiplication: Vectors (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 12 minutes, students complete a side-by-side Model Comparison on a provided two-column template. On the LEFT column (Lesson 1 Model), they reproduce or describe their original Lesson 1 prediction drawing. On the RIGHT column (Lesson 8 Model), they draw their final, complete vector model: origin at Kisumu, engine vector [20, 0] km/h drawn due east, current vector [0, −8] km/h drawn southward from the tip, resultant vector drawn from origin to the final landing point with magnitude ≈ 21.5 km/h at bearing ≈ 111.8°, and a second diagram showing the corrective heading at bearing ≈ 066.4°. Students label every element using correct mathematical vocabulary. They then write a single powerful sentence at the bottom: 'The mathematics of vectors gives the captain the power to...' The lesson closes with 3 volunteer students reading their final sentence aloud. Teacher closes with a brief, warm acknowledgement of the unit's journey.	Distribute the Model Comparison template. Say: 'On the left — your Lesson 1 model. Reproduce it or describe it. On the right — draw your final, complete vector model of the ferry problem, fully labelled. You have 8 minutes.' Set timer. Circulate and quietly affirm strong labelling: 'I love that you labelled the resultant magnitude — that is expert vector work.' After 8 minutes: 'At the bottom of your template, finish this sentence: The mathematics of vectors gives the captain the power to... Write it. Make it powerful.' Allow 2 minutes. Cold-call 3 students to read their sentences aloud — choose one strong student, one mid-level student, one who struggled earlier in the unit to demonstrate their growth. After all three have shared, say: 'Look at the left column and look at the right column. That gap — that is eight lessons. That is the power of learning. The ferry no longer drifts. The captain has the mathematics. You gave it to him.' Collect Model Comparison templates as a unit culminating artefact. Formally close the DQB by saying: 'Driving	Strategy: Before-and-After Model Comparison (Visible Learning). Placing the Lesson 1 and Lesson 8 models side by side makes conceptual growth concrete and visual. This is both a consolidation strategy and a motivational one — students see their own intellectual journey mapped onto paper. The final sentence-completion activates synthesis and personalisation of the unit's big idea. Reading aloud creates a communal closing ritual that honours collaborative learning.	Observable indicators: (1) Lesson 8 Model includes at minimum: two component vectors drawn tip-to-tail, a resultant vector, a magnitude value, and a direction angle or bearing. (2) The corrective heading diagram shows the engine vector angled north of east (not due east), confirming understanding of vector decomposition as a design tool. (3) The final sentence connects vector mathematics to purposeful human action (navigation, safety, precision) — not just abstract computation. Teacher uses Model Comparison templates as summative evidence of unit mastery. Students whose Lesson 8 model is indistinguishable from their Lesson 1 model (e.g., a single eastward arrow with no components) require immediate follow-up.

		Question — answered. Why did the Kisumu ferry miss Mbita? Because two vectors — engine and current — combined into a resultant that pointed south of east. How can mathematics help the captain? By decomposing, adding, and redirecting vectors with precision. Case closed.'		

D. TEACHER REFLECTION

After delivering this final lesson, reflect on the following questions: (1) Could every student in your class articulate — in their own words — why the ferry missed Mbita and how vectors solve the problem? If not, which specific concept (vector addition, Pythagoras for magnitude, direction as bearing, or corrective heading decomposition) was the sticking point, and how will you address it in a follow-up or future unit? (2) When you compared Lesson 1 and Lesson 8 models, did the growth feel meaningful to students, or did some learners seem disconnected from the journey? Consider how you frame the anchoring phenomenon at the start of future units to create deeper personal investment early. (3) Did the DQB accurately reflect the unit's conceptual arc? Were there questions that were never answered, or questions that were never asked but should have been? Use this as data for refining the DQB seeding strategy in Lesson 1 of your next unit. (4) The corrective heading calculation (Part E) is the hardest cognitive step — it requires students to think backwards from outcome to input. How many students reached this independently? Consider whether a mid-unit problem of this type (not just at the final lesson) would scaffold this reasoning better. (5) Reflect on Kenya-context relevance: did students respond with genuine interest to the Kisumu–Mbita ferry scenario? Could you deepen this in future by inviting a guest speaker (e.g., a Kenya Ferry Services officer, a Lake Victoria fisherman who navigates by stars and current, or a Kenyan air traffic controller from Wilson Airport in Nairobi) to make the vector mathematics feel even more alive and locally owned?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed their own Lesson 1 prediction drawings — simple, undirected east-pointing arrows — alongside the complete vector diagram of the ferry problem showing the engine vector [20, 0], the current vector [0, −8], and the resultant vector pointing south of east with magnitude ≈ 21.5 km/h at a bearing of ≈ 111.8°. They observed the DQB populated across 8 lessons, audited every question card with a traffic-light system, and witnessed all cards reach GREEN status through student-generated answers.
What did I learn?	Students learned that: (1) two simultaneous vectors — the engine's eastward push and the current's southward pull — combine by addition into a single resultant vector, explaining exactly why the ferry missed Mbita; (2) the magnitude of the resultant (≈ 21.5 km/h, true distance ≈ 64.4 km) is found using Pythagoras' theorem on the component vectors; (3) the drift angle (≈ 21.8° south of east, bearing ≈ 111.8°) is found using trigonometry; (4) the corrective heading the captain needs (bearing ≈ 066.4°, angled ≈ 23.6° north of east) is found by reverse-engineering the vector addition so the northward component of engine velocity exactly cancels the southward current; (5) vectors differ from scalars because direction is mathematically encoded and changes the physical outcome — pointing east is not the same as pointing east while accounting for the current.
How does this explain the phenomenon?	Students explained that the Kisumu ferry missed Mbita because the captain treated velocity as a scalar (speed only, pointing east) without accounting for the current as a second vector simultaneously acting on the vessel. The resultant of the two vectors pointed south of east, landing the ferry kilometres below Mbita. Using the full toolkit of Vectors I — column vector representation, vector addition, Pythagoras' theorem, trigonometric direction finding, and vector decomposition — the captain can calculate the precise corrective bearing that pre-compensates for any known current or wind, ensuring the ferry always arrives at the intended destination. This is why direction is not optional in describing real-world quantities: mathematics with direction saves time,

	resources, and lives.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.